Adversarial Robust Salient Object Detection in Optical Remote Sensing Images With Implicit Feature Enhancement

Heng Li, Feng Shao, *Member, IEEE*, Xiangchao Meng, *Senior Member, IEEE*, Hangwei Chen, Xiongli Chai, and Zhiyi Mo

***Abstract*—Deep neural networks (DNNs) have achieved significant progress in optical remote sensing images salient object detection (ORSI-SOD) and are widely applied to various remote sensing image analysis tasks. However, few SOD models demonstrate robust performance under adversarial perturbations, which ultimately leads to a decline in detection accuracy. Moreover, most existing defense methods inject fixed Gaussian noise globally into the image. Although such approaches are easy to implement, they have several limitations in inaccurate uncertainty estimation and neglecting the unique characteristics of local salient regions. Furthermore, existing adversarial defense research rarely addresses the challenges specific to the ORSI-SOD task, leaving a gap in effective defense strategies. To tackle these issues, we propose a novel defense method, which enhances the adversarial robustness of ORSI-SOD models through implicit feature enhancement. The algorithm first proposes a two-stage strategy of reverse local noise search and forward global noise optimization, enhancing generalization ability by implicitly enhancing features to better simulate network uncertainty. Then, the algorithm proposes a global-guided texture information enhancement (GTIE) module for low-level features and a global-guided semantics information enhancement (GSIE) module for high-level features, focusing on strengthening low-level texture information and enhancing the model's understanding of high-level contextual semantic features, respectively. This dual-module design effectively weakens the impact of adversarial noise, significantly improving the robustness and accuracy of object detection. Extensive experiments on three ORSI-SOD datasets demonstrate that our defense strategy better estimates the uncertainty, resulting in an average performance improvement of 23.2% in $F_\beta^{\max}$ and 34.1% in $E_\xi^{\max}$ across six ORSI-SOD models under five different adversarial attack methods. Our code will be released in the public repository at https://github.com/kexi0714/IFe.**

Received 15 March 2025; revised 6 June 2025; accepted 11 August 2025. Date of publication 13 August 2025; date of current version 22 August 2025. This work was supported in part by the Natural Science Foundation of China under Grant 62471263 and in part by the "Science and Technology Innovation Yongjiang 2035" Key Technology Breakthrough Plan Project in Ningbo under Grant 2024Z113. *(Corresponding authors: Feng Shao; Zhiyi Mo.)*

Heng Li, Xiangchao Meng, Hangwei Chen, and Xiongli Chai are with the Faculty of Information Science and Engineering, Ningbo University, Ningbo 315211, China (e-mail: 2311100161@nbu.edu.cn; mengxiangchao@nbu.edu.cn; 1010075746@qq.com; chaixiongli@nbu.edu.cn).

Feng Shao is with the Faculty of Information Science and Engineering, Ningbo University, Ningbo 315211, China, and also with the School of Electronic Information and Artificial Intelligence, Wuzhou University, Wuzhou 543002, China (e-mail: shaofeng@nbu.edu.cn).

Zhiyi Mo is with Guangxi Key Laboratory of Machine Vision and Intelligent Control, Wuzhou University, Wuzhou 543002, China (e-mail: zhiyim@gxuwz.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3598632

***Index Terms*—Adversarial defense, implicit feature enhancement, remote sensing images, salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) [1], which simulates human visual attention mechanisms to enable machines to automatically identify perceptually significant regions or objects within an image, has emerged as a crucial preprocessing technique in computer vision. SOD plays vital roles across various domains, including object recognition [2], image segmentation [3], and object tracking [4]. Recent advances in deep learning, especially convolutional neural networks (CNNs), have greatly improved SOD accuracy and efficiency by effectively capturing complex visual features.

Although CNN-based SOD approaches have achieved remarkable success in processing natural scene images (NSIs) [5], [6], their application to optical remote sensing images (ORSIs) introduces unique and significant challenges. ORSIs often suffer from high background complexity and low object-background contrast due to scale variation, edge blur, and illumination interference [7]. Moreover, imaging altitude and atmospheric conditions introduce noise and distortion, complicating small object detection [8]. These factors raise the demand for higher model's robustness and adaptability. Furthermore, the high annotation cost and limited dataset sizes lead to poor generalization and overfitting issues [9]. To tackle these challenges, researchers have proposed various CNN enhancement strategies, including multicontent complementation [10], adjacent context coordination [11], feature correlation [12], semantic matching and edge alignment [13], multiscale edge-embedded attention [14], and global extraction local exploration [15]. These advancements have contributed to notable progress in ORSI-SOD tasks, addressing some of the inherent difficulties in this domain.

However, the inherent vulnerability of deep learning-based SOD models makes them highly susceptible to adversarial attacks [16], [17], where even imperceptible perturbations in the input can significantly degrade detection performance. In remote sensing, research on adversarial attacks and defenses has primarily focused on image classification [18], [19], [20] and object detection tasks. In remote sensing image classification, adversarial examples have been extensively studied. For instance, Shi et al. [21] designed a modality-specific generator to craft effective perturbations, while Huang et al. [22]

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

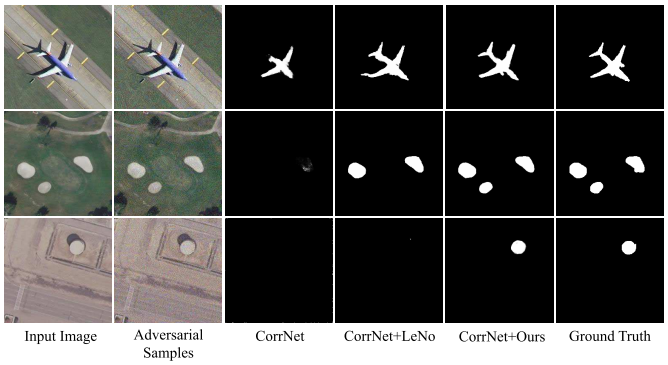


Fig. 1. Effectiveness of our proposed method. The leftmost column displays original input images, while the second column shows their corresponding adversarial examples generated by the PGD attack method. The third column presents the saliency maps of adversarial examples predicted by CorrNet. The fourth and fifth columns demonstrate output saliency maps after implementing LeNo and our defense strategy, respectively. As demonstrated in the comparative results, our defense strategy produces higher quality saliency maps with improved structural integrity and noise suppression compared to existing methods.

optimized multiple adversarial patches for stronger attacks. Similarly, in remote sensing object detection, Wang et al. [23] generated adversarial backgrounds to induce misclassification and missed detections, and Lian et al. [24] proposed a context-aware attack framework with high real-world effectiveness and transferability. As a key preprocessing step in remote sensing, the security of SOD critically affects downstream analysis. However, adversarial defense for ORSI-SOD remains underexplored [25]. As depicted in Fig. 1, adversarial perturbations can severely distort predictions, posing major risks in real-world applications. Thus, developing effective defenses is vital to enhance the model's robustness, ensure reliable interpretation of remote sensing data, and promote the integration of adversarial learning with remote sensing. This work contributes to building secure and controllable intelligent remote sensing systems.

Theoretical analysis suggests that randomization can be leveraged to enhance the robustness of deep neural networks (DNNs) [26], [27]. Building on this principle, numerous studies have explored the injection of random noise to increase network uncertainty, demonstrating significant improvements in performance. For instance, Shi et al. [25] introduced a global sinusoidal noise module after data augmentation, effectively exposing the model to a more diverse range of sample distributions. Wang et al. [28] proposed a lightweight deployment strategy by embedding a cross-shaped Gaussian noise module at the shallow feature level after the stem cell, thereby enhancing the detection performance of SOD models. Similarly, Jeddi et al. [29] incorporated Gaussian noise into each layer of the network to perturb the global feature space, employing an alternating training strategy to iteratively update both network and noise parameters.

Although the aforementioned adversarial defense methods are well-designed, they still exhibit several limitations. First, existing methods often simulate adversarial perturbations using fixed global Gaussian noise. However, the unimodal nature of the Gaussian distribution limits its ability to represent the multimodal characteristics of the real-world data. As a result, when facing adversarial samples, injecting global Gaussian noise may cause the model to converge to suboptimal solutions, thereby weakening the adversarial robustness of ORSI-SOD models and introducing practical risks. Second, most existing methods focus solely on uncertainty modeling through global noise injection, overlooking the semantic continuity and spatial correlation inherent in local salient regions of images. As a result, these approaches provide only suboptimal improvements in the model's robustness and fail to fully exploit the critical role of local salient regions in both adversarial defense and SOD. Third, current methods typically aim to improve robustness but neglect the interplay between low-level texture and high-level semantic features, as well as the model's holistic understanding of image content, which ultimately limits the model's effectiveness in target recognition tasks.

To address these limitations, this article proposes an implicit salient feature enhancement strategy to improve the adversarial robustness of ORSI-SOD networks. Specifically, to mitigate **the first limitation**, we initialize the noise using samples from a uniform distribution, enabling the model to better adapt to the complex and unknown real-world image distribution. We introduce a novel noise injection mechanism that guides the model toward more robust object detection performance. Furthermore, to help the model effectively manage the impact of noise injection and prevent instability caused by the abrupt introduction of high-intensity noise, we design a regularization mechanism that progressively increases noise intensity during training. Ultimately, we optimize both the network and noise intensity parameters using a forward global noise optimization strategy. To address **the second limitation**, we propose an inverse local noise search strategy to model the uncertainty of local salient regions. Building on this, we introduce an inverse noise patch embedding (INPE) algorithm, which identifies the locations of noise patches that induce the greatest increase in model loss and embeds local Gaussian noise blocks at these positions. This implicit strategy enhances the robustness of salient regions by injecting noise into the critical feature representations shared by the model, thereby effectively mitigating overfitting. To overcome **the third limitation**, we introduce two dedicated modules: the global-guided texture information enhancement (GTIE) module, which strengthens the model's ability to capture shallow texture features, and the global-guided semantic information enhancement (GSIE) module, which enhances its understanding of high-level contextual semantics. The major contributions of our work are summarized as follows.

1) We propose an adversarial defense algorithm with implicit feature enhancement to improve the adversarial robustness of ORSI-SOD models. This algorithm is highly compatible and can be seamlessly integrated into the existing ORSI-SOD frameworks as a plug-and-play module with minimal modifications to the original network structure.
2) We propose the inverse local noise injection and forward global noise injection strategies to effectively model the uncertainty of both global and local salient objects in images. Furthermore, we incorporate GTIE and GSIE to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

enhance the model's feature representation by strengthening low-level texture information and improving its understanding of high-level semantic content, thereby enhancing its perception of the global image context.

3) Compared to previous adversarial defense approaches in ORSI-SOD, our method offers a more comprehensive evaluation dimension by involving six state-of-the-art ORSI-SOD models. Extensive experiments on the ORSSD, EORSSD, and ORSI-4199 datasets show that our defense algorithm consistently achieves more accurate uncertainty estimation under five different attack strategies, thereby effectively improving adversarial robustness.

The remainder of this article is structured as follows. Section II provides a review of the existing adversarial attack and defense methods, while Section III introduces the proposed adversarial defense strategy. Section IV presents a discussion of the experimental results, and Section V draws the conclusion of the article.

II. Related Works

A. CNN-Based Methods for ORSI-SOD

ORSIs pose distinctive challenges for SOD owing to their high resolution, intricate scene complexity, and substantial variations in object scales and geometries. To address these challenges, CNNs have become instrumental in advancing ORSI-SOD methodologies by leveraging their hierarchical feature extraction capabilities and contextual representation learning. Recent progress in this domain has focused on architectural innovations to enhance detection precision. Benchmark datasets such as ORSSD [30] and EORSSD [7] have been established to evaluate model performance under these challenging conditions. Li et al. [30] pioneered a dual-path framework featuring an L-shaped module to aggregate complementary multiscale features from a two-stream pyramid structure, coupled with a V-shaped refinement module incorporating nested connections to preserve local object details. Concurrently, Zhang et al. [7] devised a global context-aware attention (GCA) mechanism to model long-range semantic dependencies while integrating edge-aware supervision to sharpen boundary predictions. Building upon this work, Zhou et al. [31] proposed a multiresolution encoder to fuse heterogeneous deep features extracted from three input scales, augmented by explicit edge cue integration and a hybrid loss function for saliency map optimization. In addition, Li et al. [32], who explored cross-path feature complementarity through a parallel upward fusion architecture, and Zhang and Ma [33], who introduced a weakly supervised paradigm utilizing Grad-CAM-generated pseudo labels refined via a denoising feedback network. Huang et al. [34] further advanced multiscale detection through a semantic-guided decoder architecture paired with a parallel attention fusion module for boundary refinement.

Despite the remarkable achievements of CNN-based ORSI-SOD frameworks, their practical deployment remains vulnerable to adversarial attacks. These security threats exploit imperceptible input perturbations to induce catastrophic performance degradation, exposing critical robustness deficiencies in current methodologies. The stark discrepancy between human visual perception and model susceptibility to such perturbations underscores the urgent need for defensive mechanisms. To mitigate this vulnerability, we propose a novel defense strategy designed to fortify ORSI-SOD models against adversarial noise. By strengthening the model's ability to withstand adversarial perturbations, our approach aims to improve the stability and reliability of these models in real-world scenarios, ensuring more accurate and trustworthy performance in applications where security and robustness are vital.

B. Adversarial Attack Methods

To systematically assess the effectiveness of the proposed defense mechanism, this study employs the following five representative adversarial attack strategies, each designed to challenge the model's robustness from distinct perspectives.

1) FGSM [35]: It is an adversarial attack method that applies a one-step optimization along the gradient of the loss function. Given a natural sample x and its corresponding label y, fast gradient sign method (FGSM) perturbs each element of the input x by the sign of the gradient of the loss function $J(\theta, x, y)$ with respect to the model parameters θ. The perturbation is scaled by a small hyperparameter ϵ, which controls the magnitude of the perturbation. The process for generating the FGSM adversarial sample x' is expressed as follows:

$$x' = x + \varepsilon \cdot \text{sign}\left(\nabla_x J(\theta, x, y)\right) \tag{1}$$

where $\nabla_x J(\theta, x, y)$ and sign$(\cdot)$ denote the gradient of the loss with respect to the input x, and the element-wise sign function, respectively.

2) BIFGSM [36]: It is an enhancement of the FGSM that performs multiple iterations to refine the adversarial perturbation. Unlike FGSM, which applies a single-step perturbation, basic iterative fast gradient sign method (BIFGSM) iteratively updates the input image by adding small gradients, progressively increasing the impact of perturbation. This iterative approach results in more potent adversarial examples that are more effective in fooling deep learning models. The update rule for BIFGSM is given by

$$x_{t+1} = x_t + \alpha \cdot \text{sign}\left(\nabla_{x_t} J\left(\theta, x_t, y\right)\right) \tag{2}$$

where x_t is the image at iteration t (with x_0 being the original input image), α is the step size for each iteration, $\nabla_{x_t} J(\theta, x_t, y)$ is the gradient of the loss function J with respect to the input x_t, and x_{t+1} is the updated image after applying the perturbation.

3) MIFGSM [37]: It is an extension of the BIFGSM that incorporates momentum to accelerate the optimization process. By introducing a momentum term, momentum iterative fast gradient sign method (MIFGSM) smooths out the updates, which helps to avoid oscillations and allows for larger perturbations to be generated more effectively. This method accumulates the gradient information over iterations, guiding the perturbation more efficiently in the direction that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

maximizes the loss, thereby producing stronger adversarial examples. The update rule for MIFGSM is given by

$$v_{t+1} = \beta v_t + \frac{\nabla_{x_t} J(\theta, x_t, y)}{\left\| \nabla_{x_t} J(\theta, x_t, y) \right\|_1}, \quad x_{t+1} = x_t + \alpha \cdot \operatorname{sign}\left(v_{t+1}\right) \tag{3}$$

where v_t is the momentum term at iteration t and β is the momentum coefficient.

4) PGD [38]: It is one of the most widely used and powerful adversarial attack methods. Projected gradient descent (PGD) extends the BIFGSM by allowing for multiple iterations, similar to BIFGSM, but with the added feature of projecting the perturbed image back into a valid domain after each update. This projection ensures that the generated adversarial example remains within a predefined perturbation budget, usually bounded by the ℓ_∞-norm. PGD is considered as one of the strongest white-box attack methods because it systematically explores the input space to maximize the adversarial impact. The update rule for PGD is given by

$$x_{t+1} = \operatorname{Proj}_{\mathcal{X}}\left(x_t + \alpha \cdot \operatorname{sign}\left(\nabla_{x_t} J(\theta, x_t, y)\right)\right) \tag{4}$$

where $\operatorname{Proj}_{\mathcal{X}}(\cdot)$ denotes the projection operator that maps the perturbed input back into the set $\mathcal{X}$ (the domain of valid inputs, typically constrained by a norm bound).

5) APGD [39]: It is a more recent and advanced variant of PGD designed to automatically adapt the step size and improve the efficiency of the attack. Unlike standard PGD, which uses a fixed step size throughout the iterations, automatic projected gradient descent (APGD) dynamically adjusts the step size based on the model's feedback, allowing for faster convergence and more effective adversarial perturbations. This adaptive nature helps to avoid redundant updates and enhances the attack's ability to bypass defenses. APGD is particularly effective in scenarios where a robust defense is in place, as it can fine-tune the perturbations more precisely. The update rule for APGD is given by

$$x_{t+1} = \operatorname{Proj}_{\mathcal{X}}\left(x_t + \alpha_t \cdot \operatorname{sign}\left(\nabla_{x_t} J(\theta, x_t, y)\right)\right) \tag{5}$$

where α_t is the dynamically adjusted step size at iteration t.

C. Defense Against Adversarial Attacks

Adversarial defense strategies can be broadly categorized into model optimization, adversarial training, and input transformation-based defenses. Model optimization-based defenses primarily aim to enhance the intrinsic robustness of neural networks by designing more resilient architectures or integrating specialized modules [40], [41]. For instance, Erichson et al. [40] improved adversarial robustness by introducing JumpReLU, a modification of the traditional ReLU activation function, which effectively mitigates adversarial attacks. Similarly, Li et al. [41] enhanced the robustness of CNNs by replacing conventional max-pooling, strided convolution, and average-pooling operations with a discrete wavelet transform (DWT). By decomposing feature maps into low- and high-frequency components, DWT improves the model's ability to resist adversarial perturbations. Moreover, Dong et al. [42] proposed a random normalization aggregation to enhance adversarial defense by replacing batch normalization layers and aggregating various normalization types to form a random space, making it difficult for adversaries to transfer attacks. Adversarial training is another widely adopted defense mechanism, which strengthens the model's robustness by incorporating adversarial examples during training [43]. By exposing models to adversarial perturbations throughout the training process, adversarial training enables them to learn more resilient representations. Building upon this, various advanced adversarial training frameworks have been developed. For instance, TRADES [44] introduced a trade-off mechanism between robustness and accuracy by decomposing the total loss into natural error and boundary error. This helps to balance the model's performance on clean and adversarial examples. MART [45] improves upon standard adversarial training by giving special attention to misclassified adversarial examples, thereby enhancing the model's capacity to correctly classify perturbed inputs. Inputting transformation-based defenses aim to mitigate the effects of adversarial perturbations through preprocessing techniques that transform the input data before feeding it into the model. Common approaches include JPEG compression [46], smoothing [47], feature squeezing [48], and perturbation purifier [49]. Sun et al. [50] proposed to jointly tune adversarial exposure and additive perturbation for attack and constrain image close to cloudy image as adversarial cloud (AdvCloud). Based on this, they designed DefenseNet, a learnable preprocessing module aimed at mitigating the impact of both Vanilla AdvCloud and Generalized AdvCloud, thereby preserving the performance of ORSI-SOD models. In contrast, our work investigates the vulnerability of ORSI-SOD models under conventional adversarial attacks, such as FGSM, PGD, and their variants, rather than focusing on domain-specific perturbations like AdvCloud. More importantly, our defense strategy does not rely on access to adversarial or noisy samples during training. Instead, we propose a plug-and-play defense framework trained purely on clean data, which significantly improves the model's robustness against adversarial perturbations while maintaining practical deployment feasibility. Additionally, some studies disrupt perturbations through noise injection [25], [28], [29]. Despite the variety of existing adversarial defense strategies, most methods are specifically designed for image classification models, while limited attention has been given to defending ORSI-SOD models.

III. Proposed Method

A. Architecture Overview

In this section, we propose a new framework to improve the adversarial robustness of the original ORSI-SOD models through implicit feature enhancement. Although both theoretical and practical studies have demonstrated that randomization techniques [51] can simulate network uncertainty to enhance adversarial robustness, existing methods fail to consider the unimodal nature of Gaussian distributions, which does not fully align with the multimodal distributions of real-world image data. Additionally, the use of global noise injection alone neglects the specificity of local regions containing salient

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

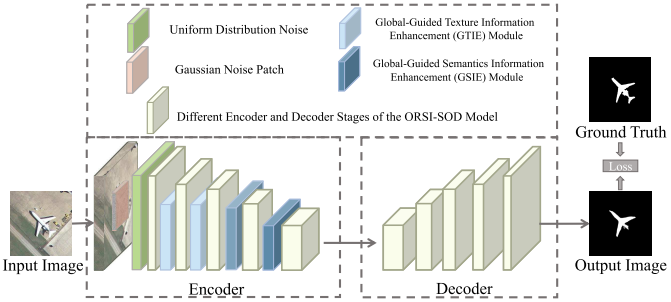


Fig. 2. Overall architecture of the proposed adversarial defense algorithm. The Gaussian noise patch and the uniform distribution noise represent the local and global noise, respectively. GTIE and GSIE are embedded into the first two and the last two main convolutional blocks during the encoding phase, respectively.

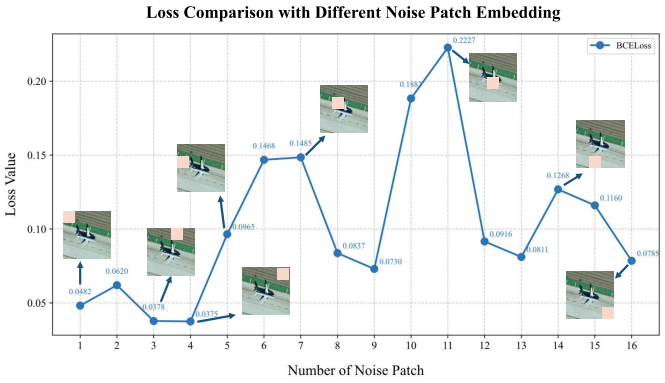


Fig. 3. Procedure of INPE. INPE iteratively injects localized noise and selects the most loss-sensitive patch for targeted perturbation, enhancing robustness in salient regions without requiring annotations.

objects. In the framework, we integrate both global and local noise injection strategies into the SOD model. Meanwhile, we embed GTIE and GSIE modules into the main convolutional blocks during the encoding phase to achieve implicit feature enhancement, thereby boosting internal model uncertainty and enhancing the saliency of detected targets. As illustrated in Fig. 2, our framework can be roughly divided into two parts: the noise injection strategy and the feature enhancement components. In the noise injection strategy, it consists of two key alternating steps.

1) *Inverse local noise search*: local gaussian noise patch is embedded into the region corresponding to the patch that cause the greatest increase in model loss using INPE. In this phase, uncertainty modeling of locally salient regions is performed through implicit feature enhancement. Only the local noise parameters are updated in this step, while other parameters are frozen.
2) *Forward global noise optimization*: global noise parameters, as well as the network and feature enhancement component weights, are updated through gradient descent, based on the global noise regularization term added to the network's loss function. During this process, the local Gaussian noise parameters remain frozen. In the feature enhancement components, we integrate GTIE and GSIE into the first two and last two main convolutional blocks during the encoding phase, respectively. This enhances the model's ability to understand image textures and semantics, reducing the likelihood of adversarial samples misleading the model. In Sections III-B and III-C, we will provide a detailed description of the components of the proposed adversarial defense algorithm.

B. Noise Injection Strategy

1) Inverse Local Noise Search: Although existing methods aim to improve the overall robustness of models, their optimization objectives are typically designed for global minimization, which may fail to sufficiently enhance feature representations in critical local regions. Consequently, salient object regions, which are often the most susceptible to adversarial perturbations, may remain inadequately protected. To address this issue, we introduce an inverse local noise search prior to the global noise optimization process. In this step, we employ a local noise injection strategy through INPE.

As shown in Fig. 3, INPE first partitions the input image into multiple patches and iteratively injects localized noise perturbations. For each patch, a noise mask is applied, and the perturbed image is evaluated using the SOD network $\mathcal{F}_\theta$. The corresponding loss is computed based on the discrepancy between the prediction of network and the ground truth. Unlike traditional approaches that minimize this loss, INPE instead identifies the patch that induces the highest loss increase, effectively pinpointing the most vulnerable region. The patch with the highest loss is then retained, and its noise distribution is updated accordingly. Rather than averaging noise perturbations across all patches, our method reinforces the most attack-sensitive regions in the ORSI-SOD model, implicitly enhancing the robustness in salient areas without explicit annotations. For the given noise patch g'', which is obtained by applying the patch embedding to the all-zero tensor g', our loss calculation can be formulated as follows:

$$loss = \frac{1}{N}\sum_{j=1}^{N}\left[\mathcal{L}\left(\mathcal{F}_\theta\left(I + g''\right), I_{gt}\right)\right] \tag{6}$$

where $\mathcal{L}$ denotes its original loss of the ORSI-SOD model.

Finally, the optimized noise patch is integrated into the original noise distribution, ensuring that the learned perturbation enhances local feature resilience. INPE introduces an inverse optimization perspective to adversarial defense, shifting from global loss minimization to max-loss patch identification. Therefore, we can identify the regions in the image most vulnerable to attacks and generate noise masks. By implicitly discovering and strengthening the features of salient object regions, INPE provides a more adaptive and effective defense mechanism. Experimental results demonstrate that INPE significantly improves the robustness of SOD networks in ORSIs, particularly under adversarial conditions. By selectively reinforcing critical regions instead of applying uniform noise optimization, INPE ensures the targeted feature protection. The complete approach is detailed in Algorithm 1.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Inverse Noise Patch Embedding (INPE)

Input: SOD network $\mathcal{F}_\theta$, patch height h, and patch width w, number of patches P, optical remote sensing images I with its ground truth I_{gt}, Gaussian noise weights g_0.

Output: Optimized Gaussian noise weights g_1.

$loss_m \leftarrow float('-inf'), \quad i_m \leftarrow None;$
for $i = 0$ **to** $P - 1$ **do**
  $g' \leftarrow \text{zeros_like}(g_0)$;
  $g'' \leftarrow \text{Patch_embed}(g', i, h, w) + g_0$;
  Compute the loss using Eq. 6;
  if $loss > loss_m$ **then**
    $i_m \leftarrow i$, $\text{loss}_m \leftarrow \text{loss}$, $g''_m \leftarrow g''$;
  end
end
Update the Gaussian noise weights g_0 to g''_m;
return g''_m

2) Forward Global Noise Optimization: To mitigate the limitations arising from the insufficient uncertainty introduced by conventional global Gaussian noise injection, we initialize global noise from a uniform distribution to compel the model to adapt to broader input perturbations. The precise control of noise injection is critical, as improper noise application may destabilize the training process. Therefore, we introduce a regularization mechanism embedded within the loss function, which supports a progressive noise intensity enhancement strategy. As the noise level gradually increases during training, the model incrementally enhances its robustness against interference, effectively preserving essential features while fully exploiting the regularization benefits conferred by noise.

Specifically, the designed noise sampling interval is $u_\epsilon \sim U\left(-\log(1+e^{\epsilon_0}), \log(1+e^{\epsilon_0})\right)$, where the dynamic parameter ϵ_0 controls noise intensity. The term $\log(1+e^{\epsilon_0})$ ensures the positivity of the noise interval. During initial training stages, a smaller ϵ_0 value is adopted to ensure stable learning of primary features, avoiding premature noise interference. As training progresses, the ϵ_0 range is systematically expanded to enhance regularization effects, preventing local optimal entrapment and improving complex feature learning. This progressive noise control strategy inherently establishes an adaptive regularization mechanism, ensuring stable convergence under varying noise intensities while effectively mitigating overfitting.

Notably, continuous growth of ϵ_0 may lead to excessive noise variance expansion, causing training instability or obscuring critical features. To address this, we introduce a $1/\epsilon_0$ regularization term in the loss function to constrain ϵ_0 growth. When ϵ_0 becomes excessively large, the gradient reduction of $1/\epsilon_0$ effectively suppresses its growth rate, preventing uncontrolled noise amplification.

This regularization term exhibits gradient-adaptive characteristics: enhanced gradients at smaller ϵ_0 values promote reasonable noise intensification, while gradient reduction at larger ϵ_0 values restricts further growth. This dynamic regulation mechanism ensures noise intensity remains within a well-regulated range, allowing the model to benefit from noise-induced regularization while maintaining training stability.

Through this design, the model efficiently learns fundamental features during training and adapts to complex tasks through controlled noise enhancement. Simultaneously, the $1/\epsilon_0$ regularization term achieves dual optimization objectives of noise intensification and training stability. This strategy not only enhances model robustness but also improves generalization across diverse scenarios, ensuring superior convergence performance under noise interference. The total loss function L_{total} of proposed adversarial defense algorithm can be formulated as follows:

$$L_{\text{total}} = \frac{1}{N}\sum_{j=1}^{N}\left[\mathcal{L}\left(\mathcal{F}_\theta(I), I_{gt}\right) + \lambda \sum_{i=1}^{K}\frac{1}{\varepsilon_{0_i}}\right] \tag{7}$$

where $\mathcal{F}_\theta(\cdot)$ represents the SOD network. K and λ refer to the feature dimension and loss weight, respectively, with λ set to $1e^{-7}$.

C. Feature Enhancement Components

Although noise injection techniques (e.g., the introduction of Gaussian noise) have been widely utilized to enhance the robustness of deep learning models, they often do not directly contribute to improving the model's ability to detect and discriminate targets, especially in the task of ORSI-SOD. Consequently, relying only on noise injection is not enough to address the complex issues brought by adversarial perturbations, which can mislead the model's predictions. To tackle this issue, we propose two complementary modules (i.e., GTIE and GSIE), aimed at enhancing the model's low-level texture understanding and high-level semantic reasoning. These modules are embedded after the main convolution blocks in the encoding stage, ensuring a more robust feature extraction process while mitigating the effects of adversarial noise. By embedding a deeper understanding of image content, these modules help to reduce the misleading influence of adversarial perturbations. As a result, the combined approach not only strengthens the model's feature representation, but also enhances its discriminative power for target detection under adversarial conditions.

1) GTIE Module: In the task of ORSI-SOD, small target detection is a significant challenge due to the highly variable scales and complex backgrounds present in remote sensing images. Traditional methods that rely on self-attention mechanisms for feature enhancement often struggle with preserving detailed spatial information [25], especially when the target size is relatively small. Although self-attention is powerful for capturing global dependencies, it may fail to emphasize crucial low-level features required for small target detection, leading to suboptimal performance in SOD tasks. To address this, we propose a multiscale cross-attention mechanism that enhances the model's ability to capture small and spatially sparse targets effectively. This mechanism not only strengthens the model's ability to capture small targets but also improves the understanding of texture information at multiple scales, allowing the model to better capture the correlation between different texture patterns.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

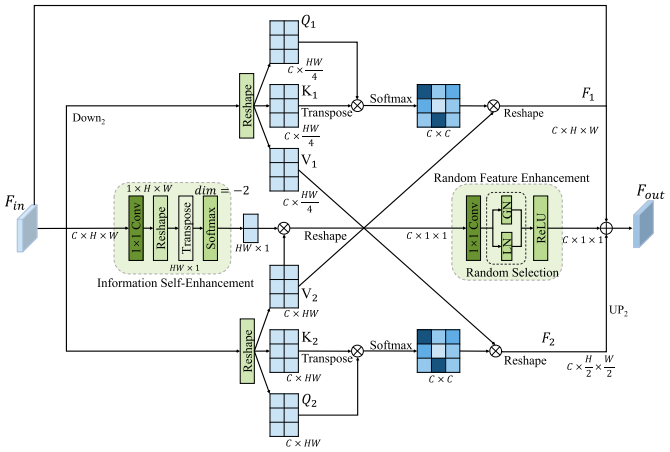


Fig. 4. GTIE module for low-level texture enhancement.

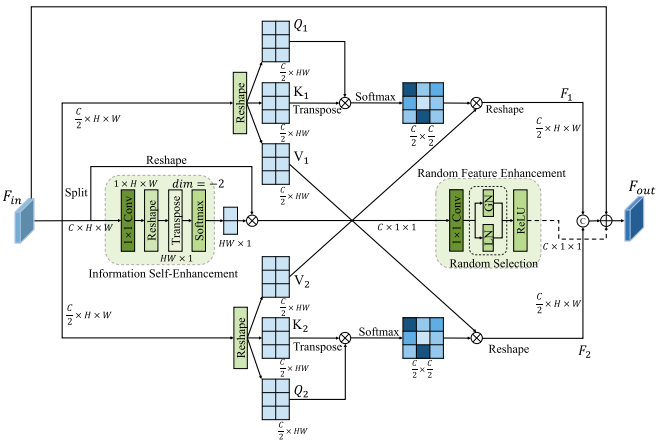


Fig. 5. GSIE module for high-level semantic refinement.

In addition, inspired by [42], we introduce a random feature enhancement mechanism in our GSIE module to improve robustness and enrich feature representations. Specifically, the module aggregates different types of normalization techniques (e.g., group normalization [52] and layer normalization [53]) to form a random space. This aggregation helps the model learn more diverse feature representations, which aids in improving its generalization and robustness against adversarial perturbations. The structural design of the GTIE is illustrated in Fig. 4.

The GTIE module processes the input feature map $\mathbf{F}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$ through a sophisticated architecture designed to enhance feature representation and capture complex dependencies within the data. The module is structured to operate on the input feature map in a multibranch fashion, where each branch contributes uniquely to the overall feature enhancement process.

The primary branch of the GTIE module begins with an information self-enhancement phase, which employs a series of operations to refine the intrinsic information of feature map. This is achieved through a 1×1 convolution followed by reshaping and transposition operations, culminating in a softmax function to generate an attention map. This attention map dynamically adjusts the weights across different channels, thereby enhancing or diminishing the relative importance of various channel features within the feature map. This process significantly improves the model's ability to perceive and differentiate between varying features. Subsequently, the random feature enhancement section integrates with the output of the information self-enhancement to further refine the feature representation and introduce stochasticity into the network. Initially, a 1×1 convolution is applied to consolidate the features. Following this, we utilize stochastic normalization aggregation as a substitute for the conventional batch normalization layer to generate a stochastic spatial domain. Finally, the ReLU activation function is employed to activate the features. The random feature enhancement component assigns a robust and effective feature expression to each spatial location, thereby enhancing the model's global perceptual capabilities. This branch of the module plays a pivotal role in augmenting the model's ability to capture and utilize texture information across the entire feature map.

In parallel, the module employs cross-attention mechanisms to model textural information across different scales, enabling interactions between fine-grained details and coarse-grained patterns. Initially, we perform downsampling and reshape operations on the input feature $\mathbf{F}_{\text{in}}$ to obtain multiscale features $\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1 \in \mathbb{R}^{C\times(HW/4)}$. Concurrently, we apply reshaping operations on $\mathbf{F}_{\text{in}}$ to derive multiscale features $\mathbf{Q}_2, \mathbf{K}_2, \mathbf{V}_2 \in \mathbb{R}^{C\times HW}$. Subsequently, matrix multiplications are executed between the transpose of $\mathbf{Q}_1$ and $\mathbf{K}_1$, and the transpose of $\mathbf{Q}_2$ and $\mathbf{K}_2$, followed by the application of the softmax function. Following this, matrix multiplications are performed across the two branches, and the results are reshaped to the original feature map size, yielding outputs from the two branches $\mathbf{F}_1 \in \mathbb{R}^{C\times H\times W}$ and $\mathbf{F}_2 \in \mathbb{R}^{C\times(H/2)\times(W/2)}$. And then, $\mathbf{F}_2$ undergoes upsampling. Ultimately, the two output features are integrated with the primary branch output and the original input features to produce the final texture-enhanced feature $\mathbf{F}_{\text{out}} \in \mathbb{R}^{C\times H\times W}$. The GTIE module processing can be expressed as

$$\mathbf{F}_1 = Soft\max\left(\mathbf{Q}_1 \otimes \mathbf{K}_1^{\mathrm{T}}\right) \otimes \mathbf{V}_2 \tag{8}$$

$$\mathbf{F}_2 = Soft\max\left(\mathbf{Q}_2 \otimes \mathbf{K}_2^{\mathrm{T}}\right) \otimes \mathbf{V}_1 \tag{9}$$

$$\mathbf{F}_{\text{out}} = \mathbf{F}_1 + \mathrm{Up}\left(\mathbf{F}_2\right) + \mathbf{F}_{\text{in}} + \mathrm{RFE}\left(\mathrm{Reshape}\left(\mathbf{V}_2 \otimes \mathrm{ISE}\left(\mathbf{F}_{\text{in}}\right)\right)\right) \tag{10}$$

where $\otimes$ and $\mathrm{Up}(\cdot)$ denote the matrix multiplication and upsampling operations, respectively. $\mathrm{RFE}(\cdot)$ and $\mathrm{ISE}(\cdot)$ represent the information self-enhancement and the random feature enhancement operations, respectively.

2) GSIE Module: In contrast to GTIE, which focuses on enhancing low-level texture information, GSIE is designed to enhance the global semantic information within the feature map. Although GTIE operates at the shallow levels of the network to capture fine-grained features like textures, it targets a more holistic representation of the scene by modeling higher level semantic information. This is achieved by utilizing cross-attention mechanisms on features split along the channel dimension, allowing the network to capture rich global dependencies across all channels. The structural design of the GTIE module is illustrated in Fig. 5.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

The feature enhancement of main branch in GSIE module is the same as GTIE. For the cross-attention branch, the GSIE module begins by splitting the original feature map into two equal parts along the channel dimension, resulting in two feature representations, each of size $(C/2) \times H \times W$. The two split feature maps are then processed separately using cross-attention operations, resulting in outputs $\mathbf{F}_1 \in \mathbb{R}^{(C/2)\times H\times W}$ and $\mathbf{F}_2 \in \mathbb{R}^{(C/2)\times H\times W}$. Finally, $\mathbf{F}_1$ and $\mathbf{F}_2$ are concatenated and fused with the primary branch output and the original features to produce the final semantics-enhanced feature $\mathbf{F}_{\text{out}} \in \mathbb{R}^{C\times H\times W}$. The entire process is formulated as follows:

$$\mathbf{F}_1 = \text{Softmax}\left(\mathbf{Q}_1 \otimes \mathbf{K}_1^{\text{T}}\right) \otimes \mathbf{V}_2 \tag{11}$$

$$\mathbf{F}_2 = \text{Softmax}\left(\mathbf{Q}_2 \otimes \mathbf{K}_2^{\text{T}}\right) \otimes \mathbf{V}_1 \tag{12}$$

$$\mathbf{F}_{\text{out}} = \text{Concat}\left(\mathbf{F}_1, \mathbf{F}_2\right) + \mathbf{F}_{\text{in}} + \text{RFE}\left(\text{Reshape}\left(\mathbf{V}_2 \otimes \text{ISE}\left(\mathbf{F}_{\text{in}}\right)\right)\right) \tag{13}$$

where $\text{Concat}(\cdot,\cdot)$ represents the concat operation.

Algorithm 2 Adversarial Defense Algorithm

Input: Optical remote sensing images I, ground truth I_{gt}, feature dimension K, initial uniform distribution noise parameter ϵ_0, and Gaussian noise weights g_0.
Parameters: The number of main convolutional blocks in the encoder B, hyperparameter λ.
Output: SOD network $\mathcal{F}_\theta$ with parameters θ_1

State that the noise u_ϵ follows a uniform distribution: $u_\epsilon \sim U(-\log(1+e^{\epsilon_0}), \log(1+e^{\epsilon_0}))$;
Sampling noise from a Gaussian distribution, which weights g_0 as 0 for all elements;
Initialize the SOD network parameters θ_0;
for $t = 0$ *to* $T-1$ **do**
  Initialized the Gaussian noise weights to g_0;
  # Reverse local noise search *(Step 1)*
  Freeze all model parameters except g_0;
  Apply u_ϵ and Gaussian noise to input feature map x and forward through the SOD model;
  for $s = 1$ *to* B **do**
    Extract features from the s-th convolutional block to obtain a feature map $\mathbf{F}_{\text{in}}$;
    Apply GTIE or GSIE to enhance feature map $\mathbf{F}_{\text{in}}$, obtain $\mathbf{F}_{\text{out}}$;
    Forward pass $\mathbf{F}_{\text{out}}$ through next convolutional block;
  end
  Optimize the Gaussian noise weights g_0 with **inverse noise patch embedding**;
  # Forward global noise optimization *(Step 2)*
  Unfreeze all model parameters except g_m'';
  Optimize the network training by minimization loss: $\frac{1}{N}\sum_{j=1}^{N}[\mathcal{L}\left(\mathcal{F}_\theta(I), I_{gt}\right) + \lambda \sum_{i=1}^{K} \frac{1}{\epsilon_i}]$;
end
return θ_1

In ORSIs, detection performance is often severely hindered by the presence of small targets, feature blurring, and adversarial noise. These challenges complicate the extraction of salient information and increase the model's vulnerability to adversarial perturbations. To address these issues, we propose a novel adversarial defense framework that integrates three key components: inverse local noise search, forward global noise optimization injection, and feature enhancement module. The inverse local noise search efficiently detects and mitigates local adversarial perturbations, while the forward global noise optimization strategy improves model robustness by injecting noise across the entire feature space during training. The feature enhancement module helps the model focus on critical target features, thereby reducing the interference from noise and improving the distinguishability of the targets in complex backgrounds. Together, these components enhance the model's generalization ability to handle diverse adversarial samples and optimize its feature representation. The combined effect of these strategies not only strengthens the model's robustness against adversarial attacks but also improves its performance in detecting salient targets. The detailed implementation process and algorithmic steps of our proposed framework are presented in Algorithm 2.

IV. Experiment Results

A. Experimental Setup

1) Datasets: We validate the proposed method on three publicly available benchmarks for ORSI-SOD task.

ORSSD [30] is the first dedicated dataset for RSI-SOD, containing 800 ORSIs with pixel-wise annotations. These images encompass diverse scenes such as ships, vehicles, aircraft, islands, and urban areas, captured under varying spatial resolutions and complex backgrounds. The dataset is partitioned into a training set of 600 images and a testing set of 200 images.

EORSSD [7] extends ORSSD to 2000 annotated ORSIs, making it the largest and most challenging RSI-SOD dataset. It introduces additional scenarios with multiscale objects, cluttered environments, and higher scene complexity. The dataset is split into 1400 training images and 600 testing images to facilitate large-scale model evaluation.

ORSI-4199 [54] is currently the largest publicly available RSI-SOD benchmark, comprising 4199 finely annotated ORSIs with pixel-level ground truths. The dataset covers a wide range of scene types, including coastal zones, agricultural fields, industrial sites, and residential areas, and the images are captured under varying spatial resolutions and complex backgrounds. To pursue fair training and large-scale evaluation, the dataset is divided into 2000 training images and 2199 testing images.

2) Evaluation Metrics: We adopt four complementary metrics that capture distinct aspects of prediction quality, including maximum F-measure ($F_\beta^{\max}$, $\beta^2 = 0.3$) [55], mean absolute error (MAE, $\mathcal{M}$), S-measure (S_α, $\alpha = 0.5$) [56], and maximum E-measure ($E_\xi^{\max}$) [57], to comprehensively assess the impact of our defense method and other compared methods on the robustness of the SOD model. Specifically, **F-measure** is the harmonic mean of precision and recall, providing a balanced assessment of the model's ability to identify relevant data while minimizing false positives and false negatives.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

TABLE I

PERFORMANCE OF SOD MODELS UNDER ADVERSARIAL ATTACKS. THE EVALUATION INCLUDES $F_\beta^{\max}$ (LARGER IS BETTER), $\mathcal{M}$ (SMALLER IS BETTER), S_α (LARGER IS BETTER), AND $E_\xi^{\max}$ (LARGER IS BETTER)

Dataset	Attack	CorrNet				MCCNet				ACCoNet				SeaNet				MEANet				GeleNet			
		$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$
ORSSD	Clean Image	.9045	.0114	.9331	.9744	.9147	.0092	.9426	.9801	.9134	.0093	.9413	.9791	.8865	.0122	.9211	.9669	.9042	.0100	.9359	.9786	.9211	.0092	.9452	.9832
	FGSM	.7816	.0301	.8427	.8985	.8344	.0209	.8858	.9322	.8551	.0170	.9038	.9336	.8137	.0233	.8722	.9259	.8216	.0279	.8753	.9248	.8983	.0122	.9292	.9723
	BI-FGSM	.6107	.0436	.7425	.8061	.7625	.0283	.8458	.9050	.7905	.0219	.8641	.9021	.7776	.0283	.8503	.9119	.7828	.0283	.8505	.9136	.8660	.0163	.9069	.9524
	MI-FGSM	.3055	.0742	.5761	.6314	.5941	.0522	.7167	.7758	.5750	.0413	.7316	.8026	.6277	.0510	.7353	.8097	.6319	.0481	.7362	.7912	.7336	.0394	.8147	.8628
	PGD	.3393	.0671	.5832	.6738	.6124	.0466	.7315	.7922	.5485	.0522	.7080	.7927	.6665	.0526	.7503	.8441	.6390	.0529	.7376	.7844	.7563	.0401	.8178	.8724
	APGD	.5242	.0525	.6811	.7605	.6982	.0379	.8063	.8431	.7263	.0284	.8312	.8659	.7169	.0398	.8117	.8662	.7221	.0380	.8109	.8724	.8062	.0261	.8687	.9123
EORSSD	Clean Image	.8661	.0094	.9212	.9590	.8859	.0073	.9341	.9731	.8767	.0085	.9292	.9728	.8625	.0080	.9199	.9674	.8763	.0079	.9284	.9707	.8874	.0078	.9347	.9792
	FGSM	.6395	.0232	.7720	.8293	.7807	.0145	.8633	.9139	.7977	.0292	.8738	.9218	.7843	.0129	.8699	.9232	.7901	.0226	.8629	.9122	.8522	.0098	.9069	.9616
	BI-FGSM	.3448	.0368	.6158	.6525	.5894	.0214	.7562	.8063	.7148	.0206	.8320	.8789	.6936	.0178	.8106	.8821	.6917	.0219	.8012	.8593	.7943	.0113	.8710	.9278
	MI-FGSM	.1674	.0433	.5445	.5890	.3385	.0350	.6215	.6785	.5646	.0212	.7464	.8096	.5254	.0284	.7028	.7632	.5024	.0312	.6873	.7279	.6646	.0259	.7723	.8170
	PGD	.1081	.0435	.5188	.6080	.2608	.0342	.5948	.6911	.5673	.0211	.7483	.8066	.5509	.0305	.7132	.7832	.4702	.0331	.6721	.7158	.6928	.0246	.7865	.8384
	APGD	.2777	.0382	.5850	.6362	.5014	.0254	.7063	.7702	.7022	.0148	.8252	.8803	.6669	.0226	.7868	.8479	.6462	.0256	.7642	.8262	.7752	.0158	.8493	.9020
ORSI-4199	Clean Image	.8449	.0422	.8494	.9247	.8641	.0332	.8704	.9381	.8634	.0370	.8759	.9361	.8593	.0324	.8673	.9384	.8654	.0344	.8679	.9398	.8809	.0287	.8823	.9500
	FGSM	.7401	.0695	.7537	.8535	.7979	.0467	.8175	.8957	.8270	.0512	.8427	.9078	.8157	.0426	.8304	.9100	.8229	.0481	.8270	.9089	.8626	.0341	.8605	.9393
	BI-FGSM	.6074	.0955	.6617	.7530	.7027	.0662	.7579	.8416	.7876	.0663	.8138	.8805	.7728	.0513	.8021	.8840	.7837	.0602	.8007	.8815	.8436	.0395	.8455	.9266
	MI-FGSM	.3993	.1251	.5388	.6143	.4817	.1157	.6176	.7027	.6617	.1424	.7157	.7706	.6593	.0747	.7188	.8132	.6643	.0894	.7199	.7936	.7825	.0540	.7960	.8823
	PGD	.2670	.1360	.4901	.5461	.3384	.2207	.5139	.5547	.6001	.2422	.6394	.6885	.6481	.0760	.7155	.8053	.6778	.0906	.7303	.8207	.8007	.0508	.8091	.8949
	APGD	.4949	.1101	.5938	.6815	.5825	.0882	.6821	.7726	.7421	.0993	.7767	.8371	.7305	.0609	.7721	.8555	.7556	.0647	.7810	.8625	.8302	.0430	.8324	.9180

MAE quantifies the average magnitude of the errors between predicted and actual values at the pixel level. **S-measure** evaluates the structural similarity between the predicted and ground-truth data, considering both regional and object-level consistency. **E-measure** combines local pixel-level matching accuracy with global image-level statistical consistency to provide a comprehensive evaluation. MAE is the smaller the better, while the other three metrics are better when they are larger. For clarity, we note that $\mathcal{M}$ is the lower the better, while $F_\beta^{\max}$, S_α, and $E_\xi^{\max}$ are the higher the better.

3) Implementation Details: We utilized five of the most representative gradient-based adversarial attack methods, namely FGSM, BIFGSM, MIFGSM, PGD, and APGD, to attack the white-box ORSI-SOD model CorrNet [12]. Using these five attack methods, we constructed five adversarial sample datasets for the ORSI-SOD models and employed them to conduct black-box attacks on five models: MCCNet [10], ACCoNet [11], SeaNet [13], MEANet [14], and GeleNet [15]. Specifically, we set the attack step size of FGSM to 0.024. For the BIFGSM attack, we set the attack step size to 0.012 and limited the number of iterations to 10. The MIFGSM attack was configured with a step size of 0.016 and five iterations. For the PGD attack, we set the step size to 0.016 and the number of iterations to 10. At last, the APGD attack was initialized with a step size of 0.016 and the maximum number of iterations was set to five. Additionally, we set the number of patches to 16 to balance computational complexity and defense effectiveness, and all training parameters for each ORSI-SOD model were set consistent with the original papers. We also applied the same data augmentation techniques to the six models, specifically by flipping and rotating the EORSSD, ORSSD, and ORSI-4199 training datasets to generate seven additional variants of the original training data. To comprehensively analyze the impact of pretrained weights on model robustness in our experiments, we adopted ResNet-50 [58] as the encoder backbone in ACCoNet. Overall, the robustness evaluation of ORSI-SOD models was structured along four key dimensions, as illustrated in Fig. 6.

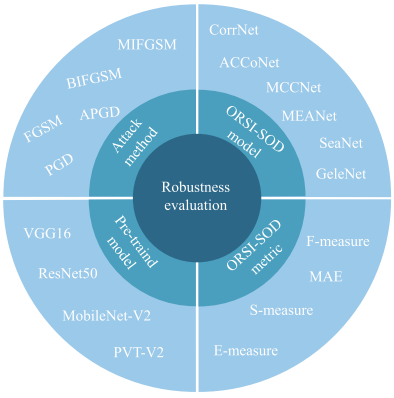


Fig. 6. Four dimensions in experimental settings.

B. Sensitivity of the Models to Adversarial Attacks

We evaluate the performance of six popular SOD networks (i.e., CoorNet, MCCNet, ACCoNet, SeaNet, MEANet, and GeleNet) on both natural images and adversarial samples generated based on the CoorNet model. As shown in Table I, all six networks exhibit varying degrees of performance degradation on the ORSSD, EORSSD, and ORSI-4199 datasets across four evaluation metrics: $F_\beta^{\max}$, $\mathcal{M}$, S_α, and $E_\xi^{\max}$. For instance, on the ORSSD dataset, the $F_\beta^{\max}$ of CoorNet decreased by 0.57 under PGD attack, and $F_\beta^{\max}$ of ACCoNet dropped from 0.9134 to 0.5485. On the EORSSD dataset, $\mathcal{M}$ of MCCNet increased from 0.0073 to 0.0350 under MIFGSM attack, while $E_\xi^{\max}$ of MEANet dropped by 0.24, and S_α of SeaNet decreased from 0.9199 to 0.7028. On the ORSI-4199 dataset, GeleNet exhibited a notable performance degradation, with $F_\beta^{\max}$ decreasing by 0.09 and $\mathcal{M}$ increasing from 0.0287 to 0.0540. The experimental results indicate that CoorNet experienced the most significant performance degradation, likely due to the adversarial samples being generated based on its gradients. The other four CNN-based ORSI-SOD models, which share structural similarities with CoorNet, also show varying degrees of performance decline. For instance, MCCNet demonstrated worse robustness than the other models on the EORSSD adversarial dataset, possibly due to the influence of the model backbone VGG-16 being consistent with the white-box model CorrNet. In contrast, the transformer-based model GeleNet experienced the least performance degradation among all the test models. This enhanced robustness is potentially attributed to its distinct architectural design compared

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

TABLE II

PERFORMANCE COMPARISON OF DEFENSE ALGORITHMS ON FGSM, BIFGSM, MIFGSM, PGD, AND APGD ADVERSARIAL SAMPLES ON THE ORSSD DATASET. THE EVALUATION INCLUDES $F_\beta^{\max}$ (LARGER IS BETTER), $\mathcal{M}$ (SMALLER IS BETTER), S_α (LARGER IS BETTER), AND $E_\xi^{\max}$ (LARGER IS BETTER). THE TOP TWO RESULTS ARE MARKED IN RED, AND BLUE, RESPECTIVELY

Attack	Method	CorrNet				MCCNet				ACCoNet				SeaNet				MEANet				GeleNet			
		$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$
FGSM	No Defense	.7816	.0301	.8427	.8985	.8344	.0209	.8858	.9322	.8551	.0170	.9038	.9336	.8137	.0233	.8722	.9259	.8216	.0279	.8753	.9248	.8983	.0122	.9292	.9723
	JPEG	.7826	.0250	.8531	.9097	.8051	.0238	.8734	.9217	.8134	.0255	.8718	.9215	.8118	.0221	.8737	.9251	.7931	.0258	.8600	.9205	.8666	.0161	.9078	.9588
	Smooth	.8027	.0247	.8589	.9165	.8524	.0177	.8962	.9459	.8715	.0160	.9083	.9496	.8243	.0210	.8828	.9241	.8437	.0203	.8930	.9422	.8860	.0130	.9232	.9693
	RNA	.8437	.0220	.8907	.9440	.8551	.0200	.8944	.9536	.8586	.0174	.9059	.9565	.7606	.0325	.8315	.9109	.8231	.0249	.8657	.9340	.8991	.0109	.9355	.9751
	LeNo	.8303	.0221	.8767	.9300	.8713	.0160	.9078	.9560	.8751	.0152	.9110	.9619	.8208	.0205	.8774	.9307	.8513	.0184	.8968	.9474	.8899	.0128	.9222	.9696
	LeWD	.8408	.0201	.8876	.9401	.8787	.0155	.9178	.9635	.8879	.0155	.9209	.9709	.8400	.0162	.8976	.9525	.8680	.0155	.9096	.9595	.8984	.0120	.9265	.9742
	Ours	.8578	.0181	.8985	.9502	.8861	.0150	.9238	.9710	.9006	.0124	.9308	.9798	.8791	.0118	.9178	.9742	.8846	.0126	.9224	.9716	.9069	.0111	.9307	.9788
BI-FGSM	No Defense	.6107	.0436	.7425	.8061	.7625	.0283	.8458	.9050	.7905	.0219	.8641	.9021	.7776	.0283	.8503	.9119	.7828	.0283	.8505	.9136	.8660	.0163	.9069	.9524
	JPEG	.7586	.0268	.8404	.9003	.7841	.0261	.8600	.9121	.7902	.0243	.8571	.9139	.7940	.0235	.8641	.9206	.7893	.0269	.8580	.9165	.8616	.0153	.9064	.9554
	Smooth	.7388	.0308	.8274	.8712	.8271	.0215	.8821	.9350	.8494	.0178	.8973	.9356	.8167	.0220	.8761	.9199	.8259	.0245	.8779	.9321	.8715	.0137	.9134	.9604
	RNA	.8070	.0253	.8630	.9212	.8132	.0243	.8634	.9248	.8434	.0209	.8937	.9510	.7464	.0334	.8231	.8955	.7841	.0312	.8375	.9136	.8741	.0136	.9141	.9626
	LeNo	.7968	.0230	.8558	.9095	.8557	.0168	.8985	.9469	.8374	.0182	.8928	.9453	.8056	.0221	.8671	.9185	.8150	.0214	.8757	.9232	.8804	.0129	.9156	.9650
	LeWD	.8016	.0209	.8741	.9203	.8647	.0160	.9072	.9564	.8466	.0180	.8968	.9505	.8153	.0174	.8757	.9253	.8266	.0185	.8970	.9317	.8923	.0122	.9231	.9712
	Ours	.8463	.0188	.8923	.9448	.8736	.0152	.9158	.9658	.8958	.0125	.9308	.9777	.8730	.0126	.9145	.9720	.8782	.0133	.9183	.9662	.9041	.0114	.9306	.9774
MI-FGSM	No Defense	.3055	.0742	.5761	.6314	.5941	.0522	.7167	.7758	.5750	.0413	.7316	.8026	.6277	.0510	.7353	.8097	.6319	.0481	.7362	.7912	.7336	.0394	.8147	.8628
	JPEG	.5568	.0529	.6892	.7737	.6629	.0430	.7773	.8529	.6518	.0387	.7729	.8174	.6721	.0353	.7874	.8524	.6553	.0561	.7673	.8295	.8030	.0274	.8594	.9122
	Smooth	.5555	.0516	.6943	.7682	.6972	.0370	.7989	.8651	.7195	.0291	.8197	.8655	.7144	.0343	.8069	.8739	.7132	.0408	.8006	.8729	.8184	.0240	.8746	.9358
	RNA	.6941	.0419	.7815	.8528	.6974	.0412	.7776	.8511	.7700	.0316	.8451	.9069	.6664	.0491	.7561	.8619	.7047	.0439	.7794	.8673	.7907	.0243	.8576	.9081
	LeNo	.6033	.0486	.7338	.7884	.7362	.0309	.8255	.8734	.6936	.0351	.8018	.8731	.7111	.0333	.8031	.8766	.7176	.0361	.7979	.8684	.8161	.0228	.8702	.9350
	LeWD	.6238	.0465	.7522	.8003	.7511	.0278	.8409	.8927	.7118	.0299	.8236	.8905	.7335	.0279	.8281	.8992	.7407	.0306	.8217	.8944	.8374	.0199	.8978	.9504
	Ours	.7964	.0247	.8530	.9184	.8288	.0200	.8859	.9326	.8719	.0166	.9152	.9664	.8552	.0155	.8987	.9638	.8338	.0207	.8848	.9393	.8775	.0145	.9157	.9622
PGD	No Defense	.3393	.0671	.5832	.6738	.6124	.0466	.7315	.7922	.5485	.0522	.7048	.7927	.6665	.0526	.7503	.8441	.6390	.0529	.7376	.7844	.7563	.0401	.8178	.8724
	JPEG	.6579	.0409	.7547	.8366	.7211	.0358	.8180	.8763	.6697	.0322	.7874	.8353	.7081	.0317	.8074	.8741	.6990	.0489	.7977	.8624	.8217	.0225	.8769	.9262
	Smooth	.6693	.0431	.7587	.8204	.7552	.0300	.8347	.8926	.7464	.0277	.8375	.8803	.7431	.0338	.8261	.8914	.7554	.0340	.8303	.8918	.8366	.0207	.8862	.9381
	RNA	.7514	.0354	.8221	.8909	.7442	.0373	.8076	.8747	.8120	.0247	.8730	.9243	.6889	.0478	.7727	.8657	.7092	.0457	.7870	.8633	.7897	.0255	.8611	.9027
	LeNo	.7550	.0277	.8291	.8777	.8091	.0226	.8721	.9167	.6958	.0393	.8047	.8589	.7515	.0292	.8309	.9026	.7507	.0331	.8270	.8819	.8173	.0221	.8764	.9287
	LeWD	.7766	.0231	.8519	.8980	.8302	.0176	.8978	.9365	.7263	.0317	.8354	.8897	.7723	.0209	.8594	.9310	.7810	.0284	.8561	.9020	.8357	.0177	.8990	.9451
	Ours	.8276	.0206	.8751	.9310	.8735	.0151	.9155	.9585	.8959	.0123	.9285	.9768	.8709	.0128	.9124	.9695	.8483	.0178	.8984	.9413	.8895	.0131	.9219	.9679
APGD	No Defense	.5242	.0525	.6811	.7605	.6982	.0379	.8063	.8431	.7263	.0284	.8312	.8659	.7169	.0398	.8117	.8662	.7221	.0380	.8109	.8724	.8062	.0261	.8687	.9123
	JPEG	.7117	.0299	.8126	.8696	.7617	.0299	.8470	.8961	.7497	.0271	.8359	.8868	.7703	.0263	.8469	.9106	.7513	.0376	.8337	.8916	.8523	.0176	.8975	.9493
	Smooth	.7076	.0370	.7955	.8461	.7967	.0237	.8657	.9110	.8082	.0213	.8767	.9054	.7857	.0276	.8570	.9097	.7891	.0305	.8587	.9106	.8665	.0155	.9071	.9553
	RNA	.7833	.0301	.8438	.9053	.7917	.0267	.8524	.9075	.8250	.0230	.8804	.9370	.7225	.0380	.8060	.8858	.7475	.0383	.8150	.8874	.8390	.0188	.8911	.9377
	LeNo	.7757	.0248	.8498	.8933	.8409	.0188	.8914	.9385	.7832	.0244	.8576	.9100	.7843	.0248	.8585	.9053	.7926	.0270	.8584	.9132	.8664	.0149	.9049	.9565
	LeWD	.8080	.0219	.8686	.9182	.8566	.0167	.9187	.9492	.8192	.0187	.8737	.9431	.8276	.0190	.8840	.9359	.8316	.0209	.8857	.9379	.8838	.0131	.9170	.9658
	Ours	.8403	.0190	.8874	.9430	.8722	.0152	.9172	.9598	.8952	.0129	.9297	.9761	.8708	.0131	.9094	.9665	.8705	.0148	.9129	.9626	.9011	.0113	.9290	.9750

to CNN-based models, which may reduce its vulnerability to adversarial perturbations generated using CNN-derived gradients. Overall, these results highlight that adversarial samples lead to significant performance degradation in SOD models, which in turn compromises the reliability and practicality of such systems.

C. Adversarial Defense Compare With the State-of-the-Art

To evaluate the performance of our method, we conducted experiments on six state-of-the-art ORSI-SOD models, namely CorrNet, MCCNet, ACCoNet, SeaNet, MEANet, and GeleNet. We compared our approach with other defense methods, specifically JPEG [46], Smooth [47], RNA [42], LeNo [28], and LeNo + WDeco [59] (LeWD). Note that the WDeco defense method was originally designed for classification models. To ensure the effectiveness of defense methods in SOD tasks, we combined LeNo and WDeco as a comparative method. Tables II–IV present the detection performance of different models under various defense strategies against FGSM, BIFGSM, MIFGSM, PGD, and APGD attacks. As shown in the tables, our defense method achieves significant performance improvements across different models. It demonstrates absolute superiority across multiple evaluation metrics, excelling in maintaining high detection accuracy, reducing error rates, enhancing structural similarity, and improving perceptual consistency. For instance, on the ORSSD dataset, $F_\beta^{\max}$ and S_α is 0.102 and 0.072 higher than the second-placed method in CorrNet under MIFGSM attack, respectively. On the EORSSD dataset, $F_\beta^{\max}$ and $E_\xi^{\max}$ is 0.034 and 0.021 higher than the second-placed method in SeaNet under BIFGSM attack, respectively. Meanwhile, on the ORSI-4199 dataset, $\mathcal{M}$ and S_α is 0.003 and 0.024 higher than the second-placed method in MEANet under PGD attack, respectively. Additionally, Fig. 7 provides visual comparisons illustrating the effectiveness of our method in different remote sensing scenarios. These results strongly validate the robustness of our approach in mitigating adversarial perturbations and enhancing the resilience of SOD models in remote sensing applications, further underscoring its effectiveness and practical value.

Furthermore, to comprehensively assess the effectiveness of our defense strategy, we evaluate its performance on natural (clean) samples, as presented in Table V. Although the results indicate that the "No Defense" setting achieves the highest accuracy on clean samples, this is expected since no additional perturbation or regularization is introduced. In contrast, our method introduces carefully designed noise to improve adversarial robustness, which inevitably increases the randomness within the network and may lead to a slight decrease in clean sample performance. Nevertheless, our method still maintains relatively high detection accuracy on clean samples while significantly enhancing robustness against adversarial attacks. This balance between natural accuracy and robustness is essential for practical applications, where models must remain reliable under both clean and adversarial conditions. The consistently strong performance across different models and datasets further demonstrates the effectiveness and practicality of our approach.

D. Ablation Studies

In this section, we conduct extensive ablation experiments on the ORSSD and EORSSD datasets to validate the effectiveness of each component in the proposed defense framework, namely: 1) forward global noise optimization; 2) inverse local noise search; and 3) GTIE module and global-guided semantics information enhancement (GSIE) module. Specifically, we

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

TABLE III

PERFORMANCE COMPARISON OF DEFENSE ALGORITHMS ON FGSM, BIFGSM, MIFGSM, PGD, AND APGD ADVERSARIAL SAMPLES ON THE EORSSD DATASET. THE EVALUATION INCLUDES $F_\beta^{\max}$ (LARGER IS BETTER), $\mathcal{M}$ (SMALLER IS BETTER), S_α (LARGER IS BETTER), AND $E_\xi^{\max}$ (LARGER IS BETTER). THE TOP TWO RESULTS ARE MARKED IN RED, AND BLUE, RESPECTIVELY

Attack	Method	CorrNet				MCCNet				ACCoNet				SeaNet				MEANet				GeleNet			
		$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$
FGSM	No Defense	.6395	.0232	.7720	.8293	.7807	.0145	.8633	.9139	.7977	.0292	.8738	.9218	.7843	.0129	.8699	.9232	.7901	.0226	.8629	.9122	.8522	.0098	.9069	.9616
	JPEG	.6047	.0205	.7704	.8356	.6835	.0320	.8023	.8657	.6401	.1632	.7245	.7747	.7445	.0148	.8497	.9081	.7223	.0338	.8269	.876	.8128	.0098	.8843	.9477
	Smooth	.6863	.0235	.8103	.8620	.8177	.0126	.8842	.9326	.8297	.0109	.8972	.9450	.8025	.0122	.8775	.9354	.7963	.0211	.8702	.9180	.8443	.0100	.9056	.9601
	RNA	.8010	.0149	.8744	.9291	.8108	.0130	.8787	.9334	.8307	.0114	.8979	.9475	.7702	.0155	.8537	.9234	.8061	.0136	.8755	.9325	.8486	.0087	.9087	.9593
	LeNo	.7783	.0145	.8607	.9175	.8323	.0106	.8964	.9504	.8018	.0113	.8839	.9368	.8029	.0119	.8753	.9345	.8078	.0157	.8735	.9290	.8395	.0088	.9022	.9613
	LeWD	.7914	.0132	.8835	.9289	.8450	.0099	.9027	.9492	.8123	.0096	.8922	.9555	.8184	.0101	.8898	.9450	.8270	.0108	.8843	.9397	.8469	.0070	.9116	.9637
	Ours	.8323	.0114	.8944	.9459	.8480	.0101	.9071	.9533	.8471	.0085	.9197	.9668	.8340	.0088	.8995	.9558	.8472	.0088	.9097	.9572	.8582	.0074	.9151	.9665
BI-FGSM	No Defense	.3448	.0368	.6158	.6525	.5894	.0214	.7562	.8063	.7148	.0206	.8320	.8789	.6936	.0178	.8106	.8821	.6917	.0219	.8012	.8593	.7943	.0113	.8710	.9278
	JPEG	.5407	.0241	.7324	.8009	.6321	.0317	.7770	.8399	.5942	.2212	.6738	.7280	.7175	.0161	.8332	.8916	.6904	.0394	.8057	.8603	.7951	.0106	.8746	.9358
	Smooth	.5789	.0291	.7398	.7899	.7537	.0163	.8508	.8991	.7975	.0128	.8776	.926	.7594	.0146	.8536	.9146	.7584	.0215	.8487	.9016	.8285	.0093	.8966	.9520
	RNA	.7219	.0194	.8204	.8844	.7211	.0194	.8167	.8804	.8100	.0126	.8830	.9404	.7356	.0199	.8275	.9050	.7692	.0164	.8478	.9157	.8013	.0109	.8799	.9356
	LeNo	.7218	.0180	.8296	.8859	.8013	.0130	.8796	.9303	.7319	.0153	.8378	.9009	.7517	.0145	.8471	.9153	.7499	.0193	.8371	.8984	.8268	.0094	.8954	.9567
	LeWD	.7402	.0155	.8588	.9139	.8271	.0124	.8890	.9421	.7663	.0136	.8579	.9211	.7897	.0126	.8743	.9294	.7652	.0136	.8591	.9247	.8414	.0081	.9119	.9706
	Ours	.8216	.0122	.8862	.9409	.8398	.0102	.9034	.9495	.8342	.0095	.9045	.9502	.8238	.0096	.8952	.9499	.8387	.0088	.9030	.9532	.8580	.0074	.9149	.9699
MI-FGSM	No Defense	.1674	.0433	.5445	.5890	.3385	.0350	.6215	.6785	.5646	.0212	.7464	.8096	.5254	.0284	.7028	.7632	.5024	.0312	.6873	.7279	.6646	.0259	.7723	.8170
	JPEG	.3391	.0398	.6100	.6650	.4514	.0428	.6739	.7383	.4754	.3126	.5660	.6290	.5829	.0221	.7517	.8166	.4962	.1597	.6462	.6725	.7252	.0161	.8258	.8919
	Smooth	.3147	.0380	.6048	.6500	.5549	.0276	.7324	.7881	.6430	.0195	.7899	.8525	.6210	.0235	.7624	.8294	.6109	.0274	.7473	.8158	.7466	.0161	.8376	.8980
	RNA	.5684	.0276	.7266	.787	.5366	.0298	.7037	.7537	.6846	.0192	.8025	.8669	.6548	.0253	.7684	.8555	.6629	.0252	.7696	.8490	.6649	.0215	.7849	.8389
	LeNo	.5517	.0274	.7274	.7611	.7037	.0184	.8211	.8829	.6147	.0206	.7749	.8430	.6134	.0240	.7630	.8330	.6143	.0259	.7489	.8138	.7580	.0132	.8498	.9137
	LeWD	.5845	.0212	.7569	.7926	.7383	.0121	.8583	.9053	.6440	.0144	.8021	.8658	.6329	.0211	.7892	.8618	.6449	.0223	.7709	.8354	.7899	.0110	.8737	.9381
	Ours	.7653	.0157	.8501	.9078	.7902	.0131	.8708	.9230	.7954	.0113	.8806	.9299	.7975	.0111	.8745	.9370	.7759	.0125	.8638	.9229	.8244	.0094	.8924	.9504
PGD	No Defense	.1081	.0435	.5188	.6080	.2608	.0342	.5948	.6911	.5673	.0211	.7483	.8066	.5509	.0305	.7132	.7832	.4702	.0331	.6721	.7158	.6928	.0246	.7865	.8384
	JPEG	.3824	.0284	.6322	.6998	.5034	.0316	.7036	.7564	.5376	.2703	.6170	.6816	.6089	.0208	.7689	.8299	.5169	.1792	.6502	.6761	.7582	.0144	.8446	.9089
	Smooth	.4300	.0338	.6553	.7068	.6158	.0218	.7641	.8152	.6871	.0166	.8163	.8729	.6665	.0224	.7897	.8499	.6432	.0243	.7708	.8349	.7759	.0133	.8565	.9111
	RNA	.6410	.0211	.7717	.8336	.5873	.0258	.7373	.8038	.7523	.0163	.8456	.9025	.6764	.0244	.7845	.8586	.6958	.0215	.7980	.8667	.6924	.0205	.8038	.8669
	LeNo	.7114	.0174	.8199	.8730	.7909	.0124	.8727	.9280	.6459	.0193	.7956	.8475	.6590	.0221	.7901	.8594	.6576	.0243	.7735	.8347	.7855	.0120	.8684	.9249
	LeWD	.7404	.0146	.8423	.8986	.8130	.0117	.8948	.9415	.6761	.0174	.8189	.8720	.6857	.0174	.8213	.8868	.6825	.0219	.8010	.8627	.8146	.0101	.8841	.9438
	Ours	.8021	.0137	.8716	.9278	.8332	.0103	.8968	.9433	.8332	.0093	.9036	.9527	.8192	.0096	.8894	.9477	.8081	.0113	.8854	.9342	.8418	.0079	.9062	.9577
APGD	No Defense	.2777	.0382	.5850	.6362	.5014	.0254	.7063	.7702	.7022	.0148	.8252	.8803	.6669	.0226	.7868	.8479	.6462	.0256	.7642	.8262	.7752	.0158	.8493	.9020
	JPEG	.5139	.0265	.7059	.7805	.5916	.0311	.7584	.8225	.5397	.2848	.6176	.6755	.6845	.0177	.8123	.8752	.6338	.0782	.7565	.8039	.7900	.0111	.8695	.9349
	Smooth	.5595	.0288	.7299	.7735	.7226	.0172	.8318	.8850	.7775	.0133	.8689	.9153	.7317	.0173	.8336	.8970	.7308	.0195	.8307	.8947	.8177	.0107	.8880	.9418
	RNA	.7108	.0193	.8109	.8759	.689	.0218	.7926	.8578	.7882	.0143	.8706	.9186	.7232	.0203	.8157	.8925	.7654	.0167	.8392	.9072	.7780	.0133	.8627	.9190
	LeNo	.7438	.0157	.8393	.8972	.8167	.0118	.8842	.9407	.7233	.0149	.8365	.8947	.7371	.0161	.8372	.9043	.7351	.0209	.8251	.8885	.8205	.0101	.8891	.9521
	LeWD	.7783	.0132	.8557	.9171	.8339	.0106	.9030	.9530	.7499	.0134	.8660	.9185	.7689	.0123	.8568	.9247	.7592	.0173	.8465	.9131	.8478	.0089	.9058	.9688
	Ours	.8180	.0128	.8820	.9385	.8414	.0096	.9027	.9491	.8358	.0088	.9073	.9537	.8295	.0087	.8971	.9531	.8353	.0090	.9016	.9517	.8555	.0071	.9147	.9676

TABLE IV

PERFORMANCE COMPARISON OF DEFENSE ALGORITHMS ON FGSM, BIFGSM, MIFGSM, PGD, AND APGD ADVERSARIAL SAMPLES ON THE ORSI-4199 DATASET. THE EVALUATION INCLUDES $F_\beta^{\max}$ (LARGER IS BETTER), $\mathcal{M}$ (SMALLER IS BETTER), S_α(LARGER IS BETTER), AND $E_\xi^{\max}$ (LARGER IS BETTER). THE TOP TWO RESULTS ARE MARKED IN RED, AND BLUE, RESPECTIVELY

Attack	Method	CorrNet				MCCNet				ACCoNet				SeaNet				MEANet				GeleNet			
		$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$
FGSM	No Defense	.7401	.0695	.7537	.8535	.7979	.0467	.8175	.8957	.8270	.0512	.8427	.9078	.8157	.0426	.8304	.9100	.8229	.0481	.8270	.9089	.8626	.0341	.8605	.9393
	JPEG	.7496	.0623	.7717	.8664	.7750	.0574	.8027	.8837	.7997	.0635	.8236	.8876	.7882	.0454	.8137	.8940	.7979	.0575	.8115	.8899	.8458	.0372	.8485	.9333
	Smooth	.7662	.0625	.7773	.8724	.8118	.0452	.8275	.9047	.8284	.0537	.8398	.9074	.8152	.0418	.8284	.9092	.8192	.0534	.8232	.9026	.8546	.0366	.8546	.9356
	RNA	.8235	.0432	.8311	.9120	.8372	.0399	.8420	.9228	.8317	.0403	.8466	.9182	.7931	.0519	.8082	.8992	.8152	.0469	.8250	.9087	.8626	.0330	.8606	.9402
	LeNo	.7733	.0587	.7924	.8790	.8323	.0396	.8442	.9201	.8345	.0451	.8489	.9173	.8222	.0407	.8340	.9154	.8316	.0439	.8335	.9162	.8585	.0339	.8599	.9375
	LeWD	.7815	.0552	.8167	.8826	.8381	.0401	.8490	.9241	.8429	.0429	.8555	.9219	.8262	.0396	.8375	.9189	.8370	.0428	.8422	.9249	.8631	.0336	.8623	.9399
	Ours	.8284	.0418	.8423	.9194	.8424	.0415	.8514	.9251	.8571	.0400	.8609	.9342	.8432	.0380	.8503	.9266	.8434	.0425	.8433	.9267	.8628	.0342	.8641	.9407
BI-FGSM	No Defense	.6074	.0955	.6617	.7530	.7027	.0662	.7579	.8416	.7876	.0663	.8138	.8805	.7728	.0513	.8021	.8840	.7837	.0602	.8007	.8815	.8436	.0395	.8455	.9266
	JPEG	.7157	.0718	.7437	.8418	.7500	.0631	.7871	.8668	.7847	.0704	.8123	.8759	.7700	.0489	.8011	.8822	.7819	.0642	.8001	.8770	.8375	.0400	.8424	.9265
	Smooth	.7160	.0741	.7432	.8385	.7817	.0525	.8076	.8859	.8152	.0562	.8318	.8993	.7978	.0461	.8166	.8992	.8026	.0603	.8100	.8906	.8472	.0386	.8489	.9315
	RNA	.8006	.0504	.8142	.8992	.8094	.0458	.8223	.9061	.8194	.0428	.8359	.9109	.7908	.0521	.8047	.8993	.8041	.0494	.8148	.9037	.8446	.0371	.8478	.9287
	LeNo	.7344	.0671	.7634	.8534	.8188	.0431	.8339	.9139	.8093	.0518	.8298	.9032	.8046	.0447	.8213	.9068	.8063	.0516	.8155	.8993	.8507	.0355	.8547	.9334
	LeWD	.7522	.0580	.7885	.8727	.8271	.0420	.8377	.9188	.8224	.0467	.8411	.9224	.8167	.0403	.8354	.9187	.8220	.0486	.8233	.9082	.8591	.0348	.8674	.9375
	Ours	.8215	.0435	.8359	.9153	.8315	.0444	.8418	.9191	.8514	.0410	.8547	.9320	.8384	.0391	.8451	.9251	.8349	.0456	.8367	.9216	.8581	.0352	.8600	.9382
MI-FGSM	No Defense	.3993	.1251	.5388	.6143	.4817	.1157	.6176	.7027	.6617	.1424	.7157	.7706	.6593	.0747	.7188	.8132	.6643	.0894	.7199	.7936	.7825	.0540	.7960	.8823
	JPEG	.5883	.0962	.6485	.7475	.6560	.0846	.7221	.8031	.7187	.1071	.7594	.8183	.7054	.0615	.7579	.8441	.7037	.0940	.7457	.8192	.7891	.0561	.8019	.8877
	Smooth	.5784	.1052	.6328	.7264	.6651	.0801	.7186	.8094	.7599	.0813	.7873	.8554	.7304	.0601	.7683	.8575	.7220	.0899	.7495	.8272	.8061	.0502	.8137	.9034
	RNA	.7308	.0641	.7619	.8577	.7387	.0608	.7711	.8587	.7756	.0526	.8028	.8844	.7477	.0596	.7740	.8748	.7602	.0581	.7812	.8741	.7673	.0566	.7854	.8730
	LeNo	.6151	.0897	.6762	.7662	.7509	.0588	.7871	.8671	.6979	.0999	.7496	.8204	.7255	.0603	.7656	.8607	.7223	.0690	.7545	.8414	.8142	.0454	.8244	.9072
	LeWD	.6348	.0819	.6884	.7911	.7763	.0549	.7988	.8831	.7266	.0879	.7622	.8454	.7591	.0582	.7817	.8860	.7535	.0626	.7773	.8610	.8294	.0429	.8368	.9185
	Ours	.7825	.0579	.8042	.8863	.7918	.0528	.8137	.8958	.8289	.0467	.8390	.9152	.8162	.0439	.8284	.9105	.7996	.0562	.8087	.8949	.8444	.0384	.8488	.9299
PGD	No Defense	.2670	.1360	.4901	.5461	.3384	.2207	.5139	.5547	.6001	.2422	.6394	.6885	.6481	.0760	.7155	.8053	.6778	.0906	.7303	.8207	.8007	.0508	.8091	.8949
	JPEG	.6458	.0846	.6876	.7973	.6742	.0759	.7375	.8506	.7425	.1009	.7764	.8344	.7190	.0584	.7699	.8486	.7316	.0811	.7662	.8419	.8062	.0501	.8164	.8980
	Smooth	.6449	.0956	.6729	.7851	.6663	.0819	.7208	.8061	.7829	.0719	.8050	.8724	.7442	.0567	.7801	.8645	.7605	.0708	.7803	.8605	.8255	.0449	.8291	.9171
	RNA	.7820	.0536	.7988	.8904	.7805	.0520	.8009	.8888	.8077	.0442	.8269	.9041	.7771	.0543	.7935	.8917	.7887	.0521	.8024	.8945	.7818	.0540	.7947	.8890
	LeNo	.7147	.0659	.7511	.8459	.8033	.0461	.8227	.9037	.5988	.1977	.6603	.7430	.7448	.0563	.7815	.8728	.7557	.0607	.7769	.8726	.8378	.0387	.8413	.9236
	LeWD	.7320	.0641	.7737	.8696	.8190	.0455	.8302	.9115	.6719	.0577	.8181	.8818	.7758	.0512	.8112	.8957	.7786	.0565	.7923	.8938	.8432	.0369	.8528	.9300
	Ours	.8041	.0538	.8195	.8975	.8168	.0472	.8304	.9118	.8442	.0438	.8492	.9263	.8325	.0407	.8398	.9221	.8230	.0488	.8264	.9103	.8557	.0353	.8572	.9361
APGD	No Defense	.4949	.1101	.5938	.6815	.5825	.0882	.6821	.7726	.7421	.0993	.7767	.8371	.7305	.0609	.7721	.8555	.7556	.0647	.7810	.8625	.8302	.0430	.8324	.9180
	JPEG	.7002	.0736	.7297	.8337	.7310	.0672	.7745	.8551	.7775	.0780	.8036	.8668	.7579	.0511	.7940	.8750	.7698	.0705	.7910	.8667	.8303	.0418	.8356	.9195
	Smooth	.7042	.0794	.7272	.8277	.7557	.0590	.7873	.8686	.8088	.0593	.8254	.8943	.7797	.0494	.8039	.8869	.7913	.0633	.8009	.8817	.8399	.0407	.8422	.9282
	RNA	.7954	.0515	.8075	.8974	.8023	.0467	.8164	.9025	.8152	.0434	.8320	.9094	.7846	.0535	.7997	.8968	.7986	.0499	.8105	.8989	.8241	.0434	.8295	.9149
	LeNo	.7353	.0653	.7662	.8551	.8145	.0438	.8309	.9110	.7686	.0701	.8020	.8734	.7871	.0485	.8089	.8961	.7952	.0532	.8042	.8947	.8479	.0367	.8498	.9321
	LeWD	.7502	.0613	.7867	.8820	.8255	.0433	.8397	.9182	.7936	.0644	.8240	.8979	.8099	.0416	.8228	.9132	.8185	.0501	.8287	.9115	.8565	.0354	.8589	.9415
	Ours	.8195	.0456	.8332	.9120	.8291	.0441	.8404	.9191	.8491	.0415	.8533	.9305	.8376	.0391	.8444	.9245	.8296	.0469	.8320	.9167	.8582	.0349	.8593	.9382

observe the performance of six different ORSI-SOD networks under PGD attack.

1) Effectiveness of Forward Global Noise Optimization (GL): We deployed only the local noise injection strategy on six SOD networks to examine the effectiveness of this component, with the quantitative results presented in Table VI. As shown in the table, due to the high efficiency of global uniform noise injection, the model robustness significantly improved after implementing this component. For instance, on the ORSSD dataset, $F_\beta^{\max}$ is increased from 0.3393 to 0.7650 in CorrNet, and $\mathcal{M}$ is decreased by 0.036. On the EORSSD dataset, S_α is improved by 0.124 in SeaNet, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

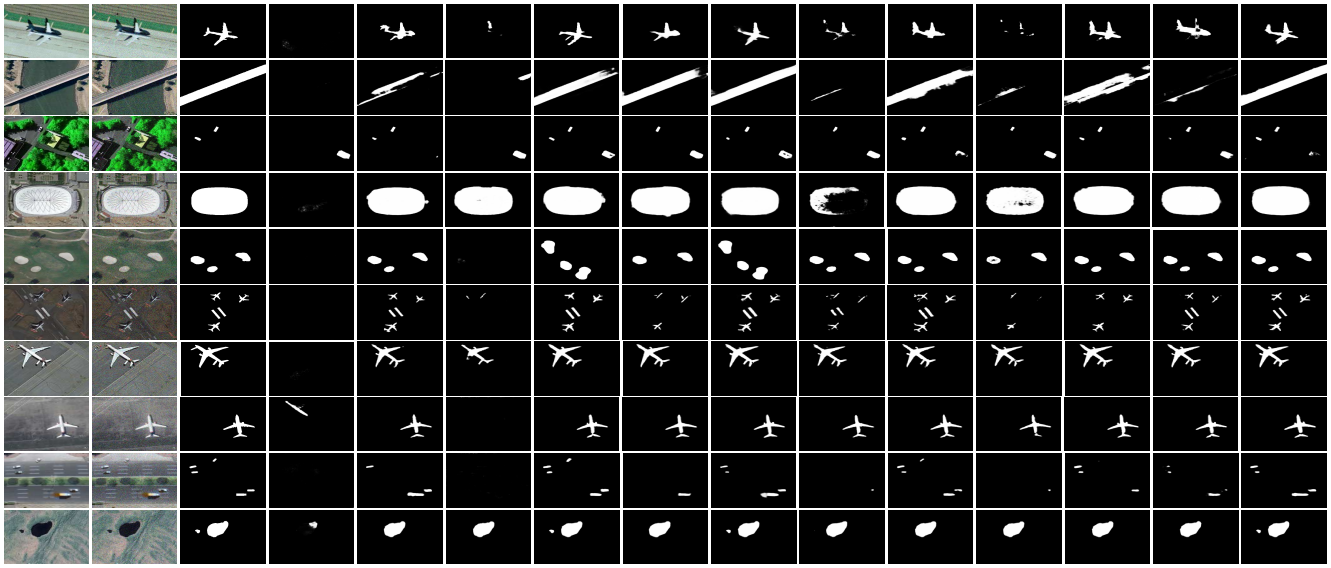


Fig. 7. Effectiveness of our proposed method. The leftmost column displays original input ORSIs, while the second column shows their corresponding adversarial examples generated by the PGD attack method. The third column presents the ground truth of salient objects. BaseNet + ours denotes the ORSI-SOD model combined with our methods.

TABLE V

PERFORMANCE COMPARISON OF DEFENSE METHODS ON NATURAL SAMPLES. THE EVALUATION INCLUDES $F_\beta^{\max}$ (LARGER IS BETTER), $\mathcal{M}$ (SMALLER IS BETTER), S_α (LARGER IS BETTER), AND $E_\xi^{\max}$ (LARGER IS BETTER). THE TOP TWO RESULTS ARE MARKED IN RED, AND BLUE, RESPECTIVELY

Dataset	Method	CorrNet				MCCNet				ACCoNet				SeaNet				MEANet				GeleNet			
		$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$
ORSSD	No Defense	.9045	.0114	.9331	.9744	.9147	.0092	.9426	.9801	.9134	.0093	.9413	.9791	.8865	.0122	.9211	.9669	.9042	.0100	.9359	.9786	.9211	.0092	.9452	.9832
	JPEG	.8692	.0155	.9103	.9549	.8902	.0121	.9228	.9678	.8808	.0142	.9174	.9582	.8685	.0152	.9042	.9602	.8719	.0153	.9109	.9600	.9064	.0109	.9345	.9770
	Smooth	.8821	.0137	.9177	.9576	.9009	.0111	.9346	.9711	.9077	.0109	.9351	.9751	.8837	.0126	.9187	.9648	.9017	.0110	.9329	.9750	.9177	.0100	.9432	.9820
	RNA	.8756	.0161	.9147	.9621	.8885	.0138	.9232	.9694	.8754	.0146	.9164	.9661	.8039	.0252	.8622	.9331	.8592	.0174	.8952	.9543	.9242	.0076	.9452	.9859
	LeNo	.8718	.0155	.9077	.9532	.8964	.0116	.9292	.9729	.8948	.0131	.9248	.9734	.8533	.0167	.8987	.9482	.8845	.0132	.9196	.9665	.9036	.0111	.9325	.9779
	LeWD	.8699	.0161	.9060	.9502	.8933	.0129	.9261	.9686	.8890	.0145	.9179	.9694	.8482	.0184	.8940	.9433	.8797	.0149	.9110	.9609	.8988	.0121	.9295	.9716
	Ours	.8769	.0155	.9127	.9584	.9027	.0114	.9344	.9769	.9058	.0114	.9376	.9805	.8773	.0119	.9175	.9704	.8948	.0115	.9263	.9745	.9122	.0108	.9377	.9808
EORSSD	No Defense	.8661	.0094	.9212	.9590	.8859	.0073	.9341	.9731	.8767	.0085	.9292	.9728	.8625	.0080	.9199	.9674	.8763	.0079	.9284	.9707	.8874	.0078	.9347	.9792
	JPEG	.8177	.0116	.8907	.9337	.8588	.0102	.9141	.9590	.8544	.0127	.9104	.9581	.8424	.0099	.9065	.9567	.8460	.0116	.9086	.9517	.8744	.0081	.9265	.9757
	Smooth	.8526	.0113	.9110	.9502	.8818	.0078	.9295	.9699	.8745	.0081	.9259	.9700	.8601	.0084	.9165	.9656	.8719	.0082	.9251	.9666	.8836	.0077	.9332	.9784
	RNA	.8631	.0100	.9193	.9620	.8783	.0072	.9299	.9694	.8645	.0094	.9157	.9664	.8068	.0124	.8827	.9413	.8465	.0096	.9070	.9542	.8870	.0062	.9351	.9808
	LeNo	.8298	.0113	.8955	.9443	.8588	.0087	.9122	.9621	.8562	.0092	.9115	.9653	.8511	.0087	.9097	.9651	.8646	.0086	.9199	.9663	.8673	.0071	.9216	.9745
	LeWD	.8235	.0141	.8907	.9388	.8546	.0100	.9059	.9579	.8514	.0107	.9051	.9591	.8466	.0118	.9010	.9549	.8595	.0112	.9138	.9542	.8634	.0089	.9169	.9683
	Ours	.8507	.0100	.9062	.9548	.8647	.0086	.9175	.9638	.8605	.0081	.9197	.9668	.8512	.0081	.9107	.9642	.8670	.0076	.9233	.9641	.8730	.0064	.9274	.9750
ORSI-4199	No Defense	.8449	.0422	.8494	.9247	.8641	.0332	.8704	.9381	.8634	.0370	.8759	.9361	.8593	.0324	.8673	.9384	.8654	.0344	.8679	.9398	.8809	.0287	.8823	.9500
	JPEG	.8260	.0465	.8357	.9143	.8433	.0401	.8547	.9220	.8555	.0414	.8693	.9298	.8452	.0360	.8557	.9286	.8501	.0394	.8542	.9287	.8742	.0306	.8757	.9472
	Smooth	.8393	.0438	.8403	.9225	.8596	.0348	.8630	.9355	.8575	.0394	.8655	.9319	.8559	.0332	.8591	.9369	.8607	.0352	.8587	.9355	.8771	.0299	.8733	.9489
	RNA	.8412	.0437	.8500	.9229	.8599	.0355	.8646	.9354	.8457	.0379	.8594	.9260	.8070	.0487	.8212	.9064	.8289	.0442	.8422	.9158	.8842	.0280	.8850	.9535
	LeNo	.8122	.0492	.8263	.9045	.8520	.0354	.8622	.9316	.8576	.0376	.8702	.9331	.8483	.0349	.8568	.9327	.8622	.0347	.8642	.9374	.8701	.0304	.8755	.9444
	LeWD	.8090	.0501	.8228	.9003	.8488	.0372	.8599	.9302	.8519	.0383	.8688	.9311	.8444	.0358	.8523	.9300	.8586	.0366	.8420	.9332	.8687	.0338	.8736	.9401
	Ours	.8391	.0401	.8510	.9257	.8548	.0391	.8631	.9320	.8637	.0380	.8681	.9385	.8553	.0351	.8629	.9337	.8595	.0378	.8600	.9358	.8693	.0322	.8728	.9440

TABLE VI

ABLATION RESULTS OF EVALUATING THE EFFECTIVENESS OF EACH COMPONENT. THE EVALUATION INCLUDES $F_\beta^{\max}$ (LARGER IS BETTER), $\mathcal{M}$ (SMALLER IS BETTER), S_α (LARGER IS BETTER), AND $E_\xi^{\max}$ (LARGER IS BETTER)

	Component	CorrNet				MCCNet				ACCoNet				SeaNet				MEANet				GeleNet			
		$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$E_\xi^{max}\uparrow$
ORSSD	No Defense	.3393	.0671	.5832	.6738	.6124	.0466	.7315	.7922	.5485	.0522	.7048	.7927	.6665	.0526	.7503	.8441	.6390	.0529	.7376	.7844	.7563	.0401	.8178	.8724
	GL	.7650	.0310	.8374	.8946	.8051	.0282	.8694	.9147	.8460	.0154	.8965	.9415	.8115	.0220	.8747	.9189	.8139	.0230	.8737	.9313	.8286	.0239	.8629	.9159
	GL + LO	.7933	.0255	.8547	.9132	.8634	.0158	.9062	.9559	.8885	.0133	.9230	.9674	.8551	.0155	.898	.9543	.8341	.0192	.8896	.9368	.8627	.0193	.9012	.9414
	GL + FE_1	.7741	.0285	.8431	.9012	.8293	.0207	.8836	.9318	.8638	.0146	.9078	.9512	.8298	.0198	.8839	.9336	.8221	.0215	.8792	.9328	.8432	.0219	.8795	.9268
	GL + FE_2	.7763	.0279	.8447	.9008	.8276	.0201	.8821	.9335	.8651	.0148	.9063	.9498	.8323	.0193	.8824	.9321	.8248	.0221	.8779	.9344	.8415	.0228	.8782	.9283
	GL + LO + FE_3	.8107	.0218	.8662	.9238	.8698	.0155	.9118	.9569	.8946	.0128	.9254	.9726	.8612	.0143	.9041	.9609	.8412	.0186	.8932	.9388	.8804	.0165	.9108	.9588
	GL + LO + FE_4	.8078	.0233	.8709	.9266	.8716	.0158	.9132	.9571	.8912	.0125	.9273	.9712	.8651	.0135	.9089	.9638	.8438	.0179	.8958	.9401	.8776	.0146	.9142	.9612
	GL + LO + FE_5 (Ours)	.8276	.0206	.8751	.9310	.8735	.0151	.9155	.9585	.8959	.0123	.9285	.9768	.8709	.0128	.9124	.9695	.8483	.0178	.8984	.9413	.8895	.0131	.9219	.9679
EORSSD	No Defense	.1081	.0435	.5188	.6080	.2608	.0342	.5948	.6911	.5673	.0211	.7483	.8066	.5509	.0305	.7132	.7832	.4702	.0331	.6721	.7158	.6928	.0246	.7865	.8384
	GL	.7058	.0197	.8175	.8756	.8038	.0125	.8808	.9287	.8067	.0114	.8846	.9306	.7910	.0117	.8738	.9319	.7822	.0156	.8579	.9242	.7937	.0198	.8449	.8975
	GL + LO	.7835	.0144	.8679	.9180	.8205	.0105	.8910	.9350	.8294	.0096	.9000	.9467	.7942	.0109	.8796	.9328	.7966	.0137	.8733	.9277	.8210	.0125	.8778	.9368
	GL + FE_1	.7396	.0175	.8379	.8958	.8102	.0118	.8879	.9312	.8183	.0107	.8903	.9371	.7935	.0114	.8757	.9323	.7874	.0149	.8671	.9256	.8036	.0178	.8573	.9172
	GL + FE_2	.7389	.0169	.8416	.8934	.8127	.0121	.8853	.9329	.8159	.0112	.8928	.9386	.7919	.0118	.8742	.9337	.7891	.0153	.8654	.9263	.8061	.0171	.8598	.9159
	GL + LO + FE_3	.7914	.0140	.8692	.9215	.8251	.0104	.8946	.9381	.8325	.0095	.9014	.9498	.8083	.0103	.8837	.9386	.8012	.0128	.8789	.9319	.8321	.0105	.8903	.9452
	GL + LO + FE_4	.7889	.0142	.8701	.9238	.8279	.0102	.8932	.9397	.8318	.0094	.9028	.9503	.8057	.0101	.8859	.9401	.8036	.0125	.8807	.9302	.8296	.0108	.8927	.9468
	GL + LO + FE_5 (Ours)	.8021	.0137	.8716	.9278	.8332	.0103	.8968	.9433	.8332	.0093	.9036	.9527	.8192	.0096	.8894	.9477	.8081	.0113	.8854	.9342	.8418	.0079	.9062	.9577

$E_\xi^{\max}$ is increased from 0.8441 to 0.9189. These results strongly validate the effectiveness of forward global noise optimization.

2) Effectiveness of Inverse Local Noise Search (LO): Building upon global noise injection, we further enhance critical feature information through local significant position

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

noise injection, which implicitly strengthens feature robustness to mitigate the negative effects of various attack methods. Additionally, it is important to note that we do not conduct an ablation study specifically for the local noise injection strategy. This is because the local noise strategy only implicitly enhances the model's perception during the training phase and does not affect the test set during the evaluation phase. As shown in Table VI, the robustness of all six models is further improved. For example, it increases $F_\beta^{\max}$ from 0.8460 to 0.8885 on the ORSSD dataset in ACCoNet and S_α from 0.8175 to 0.8679 on the EORSSD dataset in CorrNet. The combination of these two strategies enables more comprehensive protection against adversarial attacks, as the global noise injection enhances general robustness while the local noise search further fine-tunes the model's sensitivity to critical features, leading to superior performance in real-world scenarios.

3) Effectiveness of GTIE and GSIE (FE): To comprehensively evaluate the effectiveness of GTIE and GSIE, we design a series of ablation experiments comprising five configurations (FE_1 to FE_5). In FE_1, both GSIE modules are removed while GTIE is retained, and in FE_2, both GTIE modules are removed while GSIE is retained. These two settings are used to assess the individual contributions of GTIE and GSIE to robustness enhancement. In FE_3, all GSIE modules are replaced with GTIE, whereas in FE_4, all GTIE modules are replaced with GSIE. These configurations are intended to analyze the specific design differences between the two modules, particularly the effect of the multiscale feature cross-attention mechanism in GTIE on enhancing low-level texture feature representation, and the role of the global information cross-attention mechanism in GSIE in improving the perception of high-level semantic information. FE_5 corresponds to the default setting, where both GTIE and GSIE are deployed together during the encoding phase. As shown in Table VI, on the ORSSD dataset, the combination of GL with FE_1 and GL with FE_2, respectively, improves $F_\beta^{\max}$ by 0.0091 and 0.0113 compared to the baseline GL in CorrNet, demonstrating the respective effectiveness of GTIE and GSIE. Meanwhile, on the EORSSD dataset, the configurations combining GL, LO, and FE_3 and GL, LO, and FE_4 achieve $E_\xi^{\max}$ values that are 0.0125 and 0.0109 lower than that of GL, LO, and FE_5 in GeleNet, further illustrating the individual effectiveness of each component within the GTIE and GSIE modules. Moreover, the combined approach of GL, LO, and FE_5 achieves the best defense performance among all configurations, which further verifies the effectiveness of integrating both GTIE and GSIE components. Therefore, the above analysis proves that both modules contribute complementary strengths to the overall robustness of the network.

4) Inference Time: Our proposed defense strategy introduces only a slight increase in the computational complexity and inference time of the model. As shown in Table VII, although our method exhibits slightly higher FLOPs and latency compared to some existing defense techniques on certain network architectures, the additional computational cost remains minimal. More importantly, our approach achieves significantly better defense performance, clearly outperforming other methods in terms of robustness. This demonstrates that the slight increase in computational overhead is a worthwhile trade-off for the substantial gains in model security. Overall, our method maintains the practicality and efficiency required for real-world deployment while providing enhanced protection against adversarial attacks.

TABLE VII
IMPACT OF DEFENSE MECHANISMS ON COMPUTATIONAL COMPLEXITY AND LATENCY PERFORMANCE OF EACH MODEL

		CorrNet	MCCNet	ACCoNet	SeaNet	MEANet	GeleNet
FLOPs(G)	Baseline	21.31	117.15	51.33	1.81	11.33	11.69
	Baseline + LeNo	21.33	117.16	51.33	1.81	11.33	11.82
	Baseline + LeNo +WDeco	21.33	117.17	51.34	1.81	11.34	11.83
	Baseline + Ours	21.36	117.19	51.36	1.82	11.34	11.74
Latency(ms)	Baseline	27.26	28.44	39.74	31.52	49.64	40.10
	Baseline + LeNo	27.88	28.97	40.05	31.91	50.13	40.54
	Baseline + LeNo +WDeco	28.01	29.15	40.19	32.11	50.28	40.59
	Baseline + Ours	28.55	29.79	40.48	32.35	50.57	40.78

V. CONCLUSION

The adversarial robustness of ORSI-SOD models is crucial for ensuring the reliability and trustworthiness of remote sensing image analysis in real-world applications. Nevertheless, existing defense strategies largely overlook the unique requirements and challenges inherent to the ORSI-SOD task, leaving a significant gap in the development of tailored defense mechanisms. In this work, we propose a novel adversarial defense method, which enhances the adversarial robustness of ORSI-SOD models through implicit feature enhancement. Specifically, our method incorporates a two-stage noise injection strategy—comprising reverse local noise search and forward global noise optimization—to improve model generalization and more effectively simulate network uncertainty. In addition, we introduce the GTIE and GSIE modules to enhance the model's ability to handle low-level texture information and high-level semantic features. Extensive experiments validate the effectiveness of our approach, demonstrating significant improvements in both robustness and accuracy on the ORSSD, EORSSD, and ORSI-4199 datasets.

Despite these promising results, there are certain limitations in our work. First, we have not incorporated adversarial training into the model's training process, which could further enhance robustness. Additionally, the global and local noise injection strategies may lead to performance degradation on natural samples, which needs further investigation. Moreover, we have not yet considered optimization-based attack methods, which could present additional challenges for the defense mechanism. In future research, we aim to address these limitations by exploring more advanced defense strategies, integrating adversarial training, and extending our framework to counter a broader range of adversarial attacks. Through these efforts, we aspire to contribute to the development of more robust and reliable ORSI-SOD models, ultimately promoting safer and more dependable applications in remote sensing image analysis.

REFERENCES

[1] W. Wang, Q. Lai, H. Fu, J. Shen, H. Ling, and R. Yang, "Salient object detection in the deep learning era: An in-depth survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 6, pp. 3239–3259, Jun. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

[2] G. Cheng, Q. Li, G. Wang, X. Xie, L. Min, and J. Han, "SFRNet: Fine-grained oriented object recognition via separate feature refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610510.

[3] G. Li et al., "Personal fixations-based object segmentation with object localization and boundary preservation," *IEEE Trans. Image Process.*, vol. 30, pp. 1461–1475, 2021.

[4] P. Zhang, W. Liu, D. Wang, Y. Lei, H. Wang, and H. Lu, "Non-rigid object tracking via deep multi-scale spatial–temporal discriminative saliency maps," *Pattern Recognit.*, vol. 100, Apr. 2020, Art. no. 107130.

[5] Y. Luo et al., "HFMDNet: Hierarchical fusion and multilevel decoder network for RGB-D salient object detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–15, 2024.

[6] Y. Luo, F. Shao, B. Mu, H. Chen, Z. Li, and Q. Jiang, "Dynamic weighted fusion and progressive refinement network for visible-depth-thermal salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 10662–10677, Nov. 2024.

[7] Q. Zhang et al., "Dense attention fluid network for salient object detection in optical remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 1305–1317, 2021.

[8] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[9] P. Thanapol, K. Lavangnananda, P. Bouvry, F. Pinel, and F. Leprévost, "Reducing overfitting and improving generalization in training convolutional neural network (CNN) under limited sample sizes in image recognition," in *Proc. 5th Int. Conf. Inf. Technol. (InCIT)*, ChonBuri, Thailand, Oct. 2020, pp. 300–305.

[10] G. Li, Z. Liu, W. Lin, and H. Ling, "Multi-content complementation network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5614513.

[11] G. Li, Z. Liu, D. Zeng, W. Lin, and H. Ling, "Adjacent context coordination network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 526–538, Jan. 2023.

[12] G. Li, Z. Liu, Z. Bai, W. Lin, and H. Ling, "Lightweight salient object detection in optical remote sensing images via feature correlation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617712.

[13] G. Li, Z. Liu, X. Zhang, and W. Lin, "Lightweight salient object detection in optical remote-sensing images via semantic matching and edge alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5601111.

[14] B. Liang and H. Luo, "MEANet: An effective and lightweight solution for salient object detection in optical remote sensing images," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 121778.

[15] G. Li, Z. Bai, Z. Liu, X. Zhang, and H. Ling, "Salient object detection in optical remote sensing images driven by transformer," *IEEE Trans. Image Process.*, vol. 32, pp. 5257–5269, 2023.

[16] H. Li, G. Li, and Y. Yu, "ROSA: Robust salient object detection against adversarial attacks," *IEEE Trans. Cybern.*, vol. 50, no. 11, pp. 4835–4847, Nov. 2020.

[17] Z. Che et al., "Adversarial attack against deep saliency models powered by non-redundant priors," *IEEE Trans. Image Process.*, vol. 30, pp. 1973–1988, 2021.

[18] Y. Xu, B. Du, and L. Zhang, "Assessing the threat of adversarial examples on deep neural networks for remote sensing scene classification: Attacks and defenses," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1604–1617, Feb. 2021.

[19] L. Chen, Z. Xu, Q. Li, J. Peng, S. Wang, and H. Li, "An empirical study of adversarial examples on remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7419–7433, Sep. 2021.

[20] Y. Su, G. Zhang, S. Mei, J. Lian, Y. Wang, and S. Wan, "Reconstruction-assisted and distance-optimized adversarial training: A defense framework for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5624613.

[21] C. Shi et al., "Multifeature collaborative adversarial attack in multimodal remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5631815.

[22] J.-J. Huang et al., "DeMPAA: Deployable multi-mini-patch adversarial attack for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5623613.

[23] X. Wang, S. Mei, J. Lian, and Y. Lu, "Fooling aerial detectors by background attack via dual-adversarial-induced error identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5618416.

[24] J. Lian, X. Wang, Y. Su, M. Ma, and S. Mei, "CBA: Contextual background attack against optical aerial detection in the physical world," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5606616.

[25] X. Shi, Z. Li, Y. Wang, Y. Lu, and L. Zhang, "A robust adversarial defense algorithm for enhancing salient object detection in remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5631014.

[26] X. Liu, M. Cheng, H. Zhang, and C.-J. Hsieh, "Towards robust neural networks via random self-ensemble," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 369–385.

[27] X. Liu, Y. Li, C. Wu, and C.-J. Hsieh, "Adv-BNN: Improved adversarial defense through robust Bayesian neural network," 2018, *arXiv:1810.01279*.

[28] H. Wang, L. Wan, and H. Tang, "Leno: Adversarial robust salient object detection networks with learnable noise," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, vol. 37, no. 2, 2023, pp. 2537–2545.

[29] A. Jeddi, M. J. Shafiee, M. Karg, C. Scharfenberger, and A. Wong, "Learn2Perturb: An end-to-end feature perturbation learning to improve adversarial robustness," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1238–1247.

[30] C. Li, R. Cong, J. Hou, S. Zhang, Y. Qian, and S. Kwong, "Nested network with two-stream pyramid for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9156–9166, Nov. 2019.

[31] X. Zhou, K. Shen, Z. Liu, C. Gong, J. Zhang, and C. Yan, "Edge-aware multiscale feature integration network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605315.

[32] C. Li et al., "A parallel down-up fusion network for salient object detection in optical remote sensing images," *Neurocomputing*, vol. 415, pp. 411–420, Nov. 2020.

[33] L. Zhang and J. Ma, "Salient object detection based on progressively supervised learning for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9682–9696, Nov. 2021.

[34] Z. Huang, H. Chen, B. Liu, and Z. Wang, "Semantic-guided attention refinement network for salient object detection in optical remote sensing images," *Remote Sens.*, vol. 13, no. 11, p. 2163, May 2021.

[35] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[36] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," in *Artificial Intelligence Safety and Security*. Boca Raton, FL, USA: CRC Press, 2018, pp. 99–112.

[37] Y. Dong et al., "Boosting adversarial attacks with momentum," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9185–9193.

[38] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," 2017, *arXiv:1706.06083*.

[39] F. Croce and M. Hein, "Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 2206–2216.

[40] N. B. Erichson, Z. Yao, and M. W. Mahoney, "JumpReLU: A retrofit defense strategy for adversarial attacks," 2019, *arXiv:1904.03750*.

[41] Q. Li, L. Shen, S. Guo, and Z. Lai, "Wavelet integrated CNNs for noise-robust image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 7245–7254.

[42] M. Dong, X. Chen, Y. Wang, and C. Xu, "Random normalization aggregation for adversarial defense," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, Jul. 2022, pp. 33676–33688.

[43] H. Wang, Y. Deng, S. Yoo, H. Ling, and Y. Lin, "AGKD-BML: Defense against adversarial attack by attention guided knowledge distillation and bi-directional metric learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7638–7647.

[44] H. Zhang, Y. Yu, J. Jiao, E. P. Xing, L. E. Ghaoui, and M. I. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jan. 2019, pp. 7472–7482.

[45] Y. Wang, D. Zou, J. Yi, J. Bailey, X. Ma, and Q. Gu, "Improving adversarial robustness requires revisiting misclassified examples," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Oct. 2020, pp. 1–14.

[46] N. Das et al., "SHIELD: Fast, practical defense and vaccination for deep learning using JPEG compression," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2018, pp. 196–204.

[47] X. Li and F. Li, "Adversarial examples detection in deep networks with convolutional filter statistics," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Jun. 2016, pp. 5775–5783.

[48] W. Xu, D. Evans, and Y. Qi, "Feature squeezing: Detecting adversarial examples in deep neural networks," 2017, *arXiv:1704.01155*.

[49] M. Naseer, S. Khan, M. Hayat, F. S. Khan, and F. Porikli, "A self-supervised approach for adversarial robustness," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 262–271.

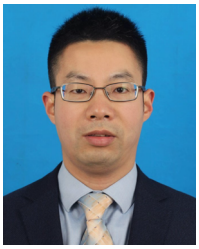
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.

[50] H. Sun et al., "Defense against adversarial cloud attack on remote sensing salient object detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Aug. 2024, pp. 8345–8354.
[51] R. Pinot et al., "Theoretical evidence for adversarial robustness through randomization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, Jul. 2019, pp. 11838–11848.
[52] Y. Wu and K. He, "Group normalization," *Int. J. Comput. Vis.*, vol. 128, pp. 742–755, Sep. 2018.
[53] J. Lei Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.
[54] Z. Tu, C. Wang, C. Li, M. Fan, H. Zhao, and B. Luo, "ORSI salient object detection via multiscale joint region and boundary model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607913.
[55] R. Achanta, S. Hemami, F. Estrada, and S. Susstrunk, "Frequency-tuned salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1597–1604.
[56] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, "Structure-measure: A new way to evaluate foreground maps," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4558–4567.
[57] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, and A. Borji, "Enhanced-alignment measure for binary foreground map evaluation," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 698–704.
[58] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 770–778.
[59] C. Zhang, Y. Li, H. Qi, and S. Lyu, "Enhancing adversarial robustness of DNNS via weight decorrelation in training," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 4660–4664.



Heng Li received the B.S. degree from Anhui Jianzhu University, Hefei, China, in 2023. He is currently pursuing the master's degree with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo, China.

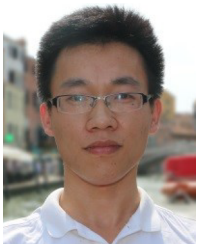
His research interests include adversarial defense and image denoising.



Feng Shao (Member, IEEE) received the B.S. and Ph.D. degrees in electronic science and technology from Zhejiang University, Hangzhou, China, in 2002 and 2007, respectively.

He is currently a Professor with the Faculty of Information Science and Engineering, Ningbo University, Ningbo, China. He was a Visiting Fellow with the School of Computer Engineering, Nanyang Technological University, Singapore, from February 2012 to August 2012. He has published over 100 technical articles in refereed journals and proceedings in the areas of 3-D video coding, 3-D quality assessment, and image perception.

Dr. Shao received the "Excellent Young Scholar" Award by NSF of China (NSFC) in 2016.



Xiangchao Meng (Senior Member, IEEE) received the B.S. degree in geographic information system from Shandong University of Science and Technology, Qingdao, China, in 2012, and the Ph.D. degree in cartography and geography information system from Wuhan University, Wuhan, China, in 2017.

He is currently a Professor with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo, China. He studied at the College of Electrical and Information Engineering, Hunan University, working as a Post-Doctoral. His research interests include multisource data fusion and machine learning.



Hangwei Chen received the B.S. degree from Ningbo University, Ningbo, China, in 2020. He is currently pursuing the Ph.D. degree in signal and information processing with Ningbo University, Ningbo, China.

His research interests include image processing, quality assessment, and visual arts.



Xiongli Chai received the B.S. degree in communication engineering, the M.S. degree in electronical and communication engineering, and the Ph.D. degree in information and communication engineering from Ningbo University, Ningbo, China, in 2017, 2020, and 2024, respectively.

He is currently a Lecturer with the Faculty of Information Science and Engineering. His research interests include image/video processing and quality assessment.



Zhiyi Mo received the Ph.D. degree from the School of Computer Science and Engineering, Guangxi Normal University, Guilin, China, in 2024.

He is a Professor with the School of Electronic Information and Artificial Intelligence, Wuzhou University, Wuzhou, China. His research interests include computer vision, target tracking, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:27:52 UTC from IEEE Xplore. Restrictions apply.